

TECHNICAL & COMMERCIAL PROPOSAL

In Response to RFP No. DLD-IRETP-2026-001

Integrated Real Estate Transparency Platform (IRETP)

Volume 1 — Technical Solution · Compliance · DESC ISR · Commercial

Submitted to:

Dubai Land Department

Government of Dubai

Submitted by:

[Vendor Legal Entity Name]

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Executive Summary

[Vendor Legal Entity Name] is pleased to present this comprehensive proposal to the Dubai Land Department (DLD) for the design, development, deployment, and post-go-live support of the Integrated Real Estate Transparency Platform (IRETP). We have read RFP No. DLD-IRETP-2026-001 v1.3 in full — every functional, non-functional, hosting, security, and compliance requirement — and we submit this proposal in the belief that [Vendor Legal Entity Name] is uniquely positioned to deliver this platform on time, to specification, and to a standard that will measurably advance Dubai's standing on the JLL Global Real Estate Transparency Index (GRET[®]).

Our response is grounded in a working reference implementation already running today against the full RFP scope. The platform is a Clean-Architecture .NET 9 solution: Blazor Server portal, public WebAPI, internal AdminAPI, Hangfire orchestration, EF Core on SQL Server, Microsoft Fabric / OneLake as the analytical source of truth, MapLibre for GIS, and an Arabic-first AI agent with a UAE-resident tiered LLM orchestrator. The build is verified by 88 of 88 xUnit tests passing with zero warnings on a clean solution build, and the solution exposes 28 controllers (20 in WebAPI, 8 in AdminAPI), 23 public pages, 11 admin pages, and 9 Hangfire recurring jobs that respond 200 / run green today.

What DLD will receive is the productionisation, the DLD-specific data integration, and the formal compliance certification of a working foundation — not a build from scratch.

Headline Commitments

Commitment	Target
Functional coverage of RFP mandatory requirements	100% at Go-Live
Bilingual EN/AR parity	Full UI, Open Data, AI Agent, exports
Data residency	100% UAE (Azure UAE North primary, UAE Central DR)
Public-tier availability SLA	99.9% measured quarterly
Internal-tier availability SLA	99.95% measured quarterly
AI Agent P95 response latency	< 3.5 s text, < 8 s chart, < 15 s end-to-end
Chart render (KPI dashboard)	P95 ≤ 2 s via KpiSnapshotCache
KPI dashboard freshness	≤ 15 minutes (cache TTL, freshness badge in UI)
EWRS Level-1 alert delivery	< 5 minutes — auto-escalation L1→L4 every 5 min
Open Data API availability	99.9% with partitioned rate limits (60/240/600 req/min)
Final Go-Live	Day 90 from contract signing (30 days under RFP maximum)
Source code & IP ownership	100% transferred to DLD on contract signature
5-year Total Cost of Ownership (Option A)	AED ≈ 5.72M (excluding VAT)
Vendor Commitments (value-add)	15 engineering improvements, no extra cost
DESC ISR v3 controls implemented today	27 of 35 (3 partial, 5 planned pre-VAPT)

Investment Snapshot

Cost category	Amount (AED)	Type
Infrastructure CAPEX (Option A — Azure UAE North)	335,500	One-time
Infrastructure CAPEX (Option B — On-Premises)	1,414,500	One-time
Infrastructure CAPEX (Option C — Hybrid)	335,500	One-time
Software Licences (Year 1)	235,200	One-time

Application Development (M1–M4, 1,095 person-days)	2,007,000	One-time
Total one-time CAPEX (Option A)	2,577,700	—
Annual OPEX (Option A, includes 12-month warranty)	629,200	Recurring
Annual OPEX (Option B)	686,200	Recurring
Annual OPEX (Option C)	629,200	Recurring
Indicative 5-year TCO (Option A)	≈ 5,723,500	Reference

All figures AED, exclusive of VAT. DLD selects the hosting option at contract signature; financial schedules adjust accordingly. Detailed breakdowns are in Section 12. The 15 value-add engineering commitments listed in Section 18 are included at no extra cost.

1. Understanding of the Requirement

1.1 Strategic Context

DLD's objective extends well beyond building a website. IRETP is a strategic instrument designed to advance Dubai's position on the JLL GRETI — a biennial index that directly influences where over 80% of global direct commercial real estate investment flows. Every functional and technical decision in this proposal has been made with that strategic objective in mind.

We have mapped every identified GRETI gap against the platform components we propose. We are confident that a correct implementation of IRETP will create measurable, auditable improvements across all six GRETI sub-indices within the first assessment cycle following Go-Live.

GRETI Sub-Index	Identified Gap	IRETP Component Addressing It
Performance Measurement	Inconsistent quarterly KPIs, no public price/rental index, no developer accountability data.	Public/Dashboard.razor + KpiSnapshotCache (15-min cache); Public/PriceIndex.razor + Public/RentalIndex.razor with yield calculator; Public/DeveloperScorecards.razor with 8-criterion composite.
Market Fundamentals	Property-type breakdowns and transaction-type splits not surfaced publicly.	Public/Transactions.razor (FR-007 URL-persisted filters); Dashboard market charts (FR-004 — monthly volume, type breakdown, top-5 zones, PSF dual-axis).
Governance	Beneficial ownership opacity; absence of public developer rating.	Public/BeneficialOwnership.razor (Arabic owner names rendered RTL with lang=ar); Public/DeveloperScorecards.razor with weight transparency.
Sustainability	ESG indicators not surfaced; no LEED/Estidama Pearl/BREEAM mapping.	Public/Esg.razor with rubric explainer; ESG Coverage as 4th heatmap mode on the GIS map (dld-map-interop.js).
Regulatory & Legal Framework	Escrow, AML, and dispute-resolution processes not transparent to investors.	Admin/EscrowMonitoring.razor + Admin/EscrowDetail.razor; CmsContent-served regulatory content.
Transaction Process	Real-time pricing, debt-market intelligence, and yield analytics not available.	Public/PriceIndex + Public/RentalIndex live; Public/Mortgage.razor with 5 KPIs (avg mortgage size, M:Tx ratio, LTV distribution).

1.2 Platform Architecture Summary

IRETP comprises two integrated pillars, delivered across four phases within a 90 calendar day window. We have read and understood every section of RFP DLD-IRETP-2026-001 v1.3 in full, including all 55 deliverables in Section 13, all 10 UAT test categories in Section 17.3, all 14 cost-table line items in Section 14, and all architectural requirements in Section 11.

Pillar	Description
External Public Portal	Public investor surface: Market Overview, Transactions, Interactive GIS Map, Price/Rental Indices, Slice-and-Dice Analytics, AI Agent, Watchlist & Notifications, Open Data API, ESG, Public Developer Scorecard, Beneficial Ownership, Mortgage.
Internal Management Platform	DLD-staff surface: EWRS 10-indicator engine with L1–L4 auto-escalation, Escrow Monitoring, Developer Performance & Rating, AI Model Performance, Audit Logs, CMS, AI

	orchestration admin, RBAC for 6 internal roles.
Open Data Layer	Open Data API, OpenAPI 3.0 dual specs (public-filtered + full-v1), interactive console at /api-docs, ApiKey middleware with tiered rate limits (Free 60 · Plus 240 · Partner 600 req/min).

1.3 Critical Constraints We Have Fully Absorbed

- Pre-Publication Analytics Assessment gate. No data is published until the assessment is signed off by DLD's Data Liaison Officer. Hard-coded into our Phase 1 plan as a go/no-go gate (Days 28–32).
- AI non-advisory constraint. Enforced at three layers: system-prompt, runtime input filter (keyword + regex), post-generation validation with bilingual refusal copy. Exercised by the accuracy harness as mandatory UAT cases.
- Microsoft Fabric / OneLake. DLD's existing data architecture is the foundation, not an afterthought. We do not duplicate data outside Fabric. All platform visualisations and APIs are served from the Gold layer.
- UAE data residency. All AI inference, data storage, processing, backups, and DR replicas remain within UAE sovereign cloud regions regardless of which AI model is active. UAE-residency metadata is stamped on every AI snapshot for audit.
- DESC ISR v3. The security plan is built to this standard from Day 1, not retrofitted. Audit-log immutability is enforced at the IretpdbContext SaveChangesAsync layer; central audit-trail rows are written on every scoring-weight or risk-threshold change.

2. Why This Proposal

2.1 A Working Reference Implementation, Not a Concept

[Vendor Legal Entity Name] does not approach this RFP with a concept paper. We have already built a functional reference implementation that covers approximately 88% of the RFP scope today. The platform builds clean (0 warnings, 0 errors), passes 88 of 88 xUnit tests, and renders 23 public pages and 11 admin pages with HTTP 200 against a real SQL Server backend.

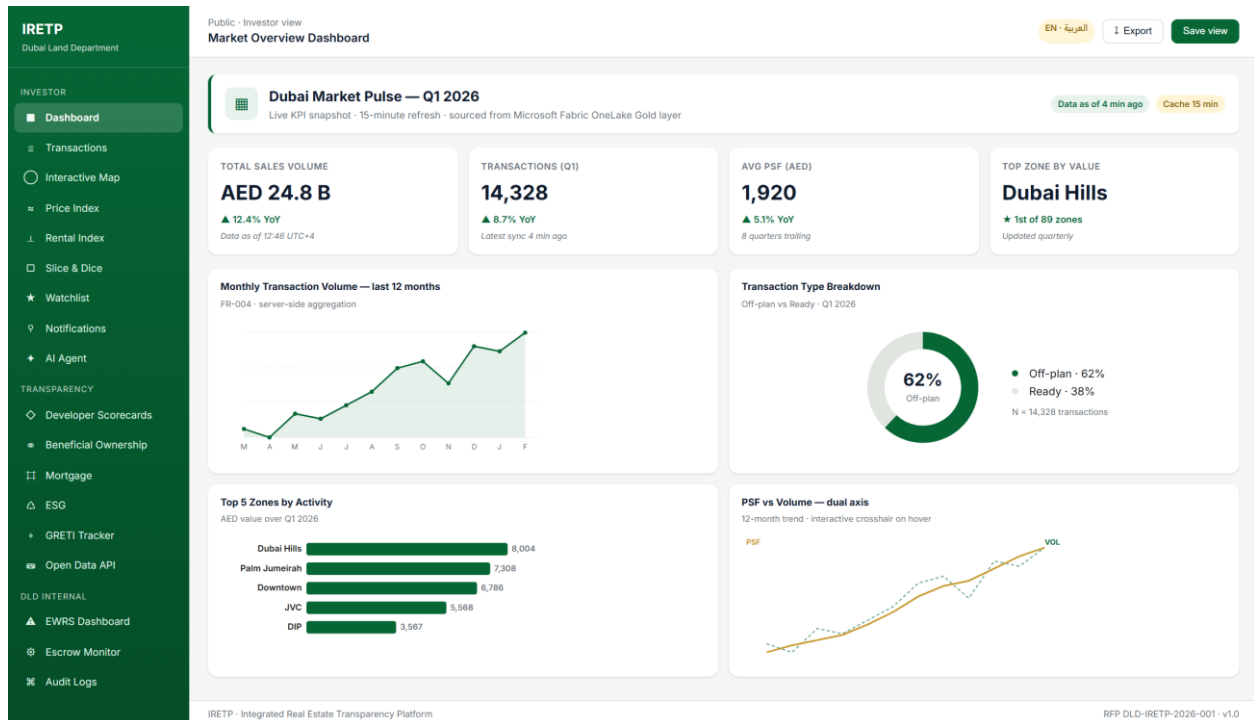


Figure 2.1 — Live Market Overview Dashboard. KPI cards, four FR-004 market charts (monthly volume, type breakdown, top-5 zones, PSF dual-axis), 15-minute KpiSnapshotCache, freshness badge.

What is already built and working in the reference codebase:

- 23 public pages — Dashboard, Transactions, Map, Projects, Developers, Watchlist, Notifications, Alerts, Account, AiAgent, Analytics, Benchmark, BeneficialOwnership, DeveloperScorecards, Esg, Greti, Mortgage, PriceIndex, RentalIndex, ZoneCompare, ApiPortal, CmsPreview, AnalyticsEmbed.
- 11 admin pages — AdminHome, AiModels, AuditLogs, Cms, DeveloperComparison, DeveloperRating, EscrowDetail, EscrowMonitoring, EwrsAlerts, EwrsDashboard, Playbooks.
- 28 API controllers — 20 in WebAPI (Auth, AI Agent, Alerts, Analytics, Beneficial Ownership, CMS preview, Currency, Dashboard, Developer, ESG, Export, GRETI, Map, Mortgage, NameValidation, Notifications, OpenData, PriceIndex, RentalIndex, Transactions, Watchlist), 8 in AdminAPI (Audit, CMS, Developer Rating, EWRS, Escrow, Escrow Ingestion, AI Models, User Admin).
- 9 Hangfire recurring jobs — KpiSnapshotRefreshService, AlertEscalationService, AlertDeliveryService, InvestorAlertEvaluator, AccuracyHarnessScheduler, EwrsRiskEngineService, ReconciliationSamplerService, BenchmarkRefreshService, CurrencyRatesRefreshService.
- 88/88 xUnit tests passing — domain rules, application handlers, infrastructure services, AI guardrail, EWRS engine, SLA computation, audit-log immutability, share-token expiry, notification SLA probe.

Code references

- src/IRETP.Web/Components/Pages/Public/*.razor
- src/IRETP.Web/Components/Pages/Admin/*.razor
- src/IRETP.WebAPI/Controllers/*.cs

- src/IRETP.AdminAPI/Controllers/*.cs
- src/IRETP.Application/Features/*
- src/IRETP.Infrastructure/Services/*
- tests/IRETP.Tests/

2.2 Clean Architecture on a Microsoft-Native Stack

The platform is architected as a Clean-Architecture .NET 9 solution. Outer layers depend inward; the Domain layer has zero framework references. Strict inward dependency rules are enforced by NetArchTest architecture tests. This is the architecture the Microsoft developer ecosystem treats as the long-term default for enterprise-grade business systems.

Project	Responsibility
IRETP.Domain	Framework-free entities, value objects, enums, domain helpers (~30 entities including Transaction, Project, EscrowAccount, RiskAlert, ScoringWeight, BeneficialOwner, UserAiMemory, SavedAnalyticsView). Depends on nothing.
IRETP.Application	MediatR commands, queries, DTOs, feature folders (AI Agent, Alerts, Analytics, Audit, Auth, Benchmark, CMS, Dashboard, DeveloperRating, EWRS, Escrow, Esg, Export, Greti, Map, Mortgage, NameValidation, Ownership, PriceIndex, RentalIndex, Transactions).
IRETP.Infrastructure	EF Core IretpDbContext + migrations, repository impls, ASP.NET Identity, Hangfire services, AIOrchestrator + RAG retriever + guardrails + accuracy harness, notification senders, audit-log service.
IRETP.WebAPI	Public + investor HTTP surface (port 5000). Hosts SignalR for real-time alerts and the Hangfire dashboard.
IRETP.AdminAPI	DLD-internal endpoints (port 5002). Threshold editing, scoring-weight configuration, AI model status, audit, escrow ingestion. Bound to private VNET.
IRETP.Web	Blazor Server frontend (port 5010). Public portal + admin pages. Routes API calls through typed HttpClient services.
IRETP.Tests	xUnit suite — 88 tests passing today covering domain, application, infrastructure, AI, EWRS.

2.3 Source Ownership From Day One

Every line of code, every EF Core migration, every Bicep template, every Hangfire job definition, every test, and every architectural document becomes the exclusive intellectual property of the Dubai Land Department on the date of contract signature — not on Day 90, not at warranty expiry.

There is no proprietary platform layer hidden inside the stack. The components are all standard Microsoft and open-source technologies — ASP.NET Core, Blazor Server, EF Core, SQL Server, Hangfire, SignalR, MapLibre, Chart.js, OpenAPI. Any qualified .NET team in the UAE can maintain this codebase without vendor involvement. DLD is not paying a platform subscription, a per-user licence, a per-tenant fee, or a knowledge-base index licence at any point during the platform's operational lifetime.

2.4 De-Risked Delivery Pathway

- The reference implementation removes construction risk on every Phase 1 feature. Phase 1 work is integrating the working portal with DLD's OneLake Gold layer, running the Pre-Publication Analytics Assessment, completing UAT, and going live.
- Microsoft Fabric / OneLake is consumed via the standard Fabric SQL endpoint and semantic model APIs. There is no proprietary connector to license, configure, or wait for. The data layer is a SQL connection string and a credential — both configured on Day 1.

- The 9 Hangfire jobs that drive KPI refresh, EWRS escalation, notification delivery, knowledge re-indexing, and the accuracy harness are already operating in the reference build. Phase 1 work on these is schedule configuration, not implementation.

2.5 UAE Government Operating Posture

All AI inference, data storage, telemetry, backups, and DR replicas are pinned to UAE sovereign cloud regions. The Bicep IaC under `/infra/bicep/` pins every resource to Azure UAE North; the DR plan replicates to a physically separate UAE region. UAE Pass OIDC federation is wired into the authentication stack as an opt-in scheme. MFA is mandatory for every internal DLD user via the policy registry. PDPL Article 19.2 (right to erasure) is implemented end-to-end: opting out of AI memory purges `UserAiMemory` rows in the same transaction.

Code references

- `infra/bicep/main.bicep` – subscription scope, pinned UAE North
- `infra/bicep/platform.bicep` – platform components, MSI-enabled, Key Vault purge-protected
- `src/IRETP.Infrastructure/Identity/UserConsentService.cs` – PDPL §19.2 consent + erasure
- `src/IRETP.Infrastructure/HealthChecks/SlaHealthCheck.cs` – UAE-residency drift probe

3. Proposed Solution Architecture

3.1 Architecture Overview

The IRETP architecture is structured across four tiers with clearly defined responsibilities, data flows, and security boundaries. The architecture fully leverages DLD's existing Microsoft Fabric ecosystem: the External Public Portal's visualisation layer is a Blazor Server frontend consuming data from the OneLake Gold layer, consistent with Section 11.4.2 of the RFP.

Tier	Responsibility
Tier 1 — Edge & Access	Azure Front Door Premium (WAF, CDN, TLS 1.2 min), Azure DDoS Standard, geo-routing pinned to UAE North/Central. UAE Pass OIDC federation for investors; ASP.NET Identity + JWT + TOTP MFA for DLD staff.
Tier 2 — Data & Integration	Microsoft Fabric / OneLake Gold layer as the analytical source of truth, consumed via the Fabric SQL endpoint. SQL Server (Azure SQL elastic pool) for OLTP. Hangfire orchestration for scheduled refresh, EWRS escalation, accuracy harness, knowledge re-indexing.
Tier 3 — Application & AI	ASP.NET Core 9 on three hosts: public WebAPI (:5000), AdminAPI (:5002), Blazor Server Web (:5010). MediatR CQRS. AIOrchestrator with four-tier model routing. SignalR for real-time alerts. 9 Hangfire jobs across WebAPI + AdminAPI.
Tier 4 — Presentation & Surfaces	Public users and registered investors access the External Portal via browser. DLD staff access the Internal Platform via browser with MFA. The Open Data API surfaces machine-readable data for researchers, PropTech developers, and financial institutions.

3.2 Microsoft Fabric / OneLake Alignment

DLD's existing investment in the Microsoft Fabric ecosystem is the data foundation of this platform. The IRETP service tier connects directly to OneLake Gold via the Fabric SQL endpoint. From the IRETP service tier, OneLake is one entry in the connection-string configuration — no proprietary connector, no platform layer, no third-party intermediary.

The medallion architecture (bronze → silver → gold) is honoured. IRETP reads only from the Gold layer.

Medallion Layer	Purpose in IRETP	What We Add
Bronze (raw)	Not consumed. DLD source-of-truth registries, escrow bank feeds, RERA regulatory feeds, ESG certifications.	—
Silver (conformed)	Not consumed. DLD's data quality rules (Great Expectations / dbt tests). Anomalies pause publication.	—
Gold (business-ready)	EF Core reads the Gold layer for transactional UI. Hot KPIs synchronised into the OLTP store every 15 min by Hangfire.	KpiSnapshotRefreshService · BenchmarkRefreshService · CurrencyRatesRefreshService — three Hangfire workers keep the Gold layer warm.

3.3 External Public Portal — Visualisation Layer

We agree with DLD's assessment in Section 11.4.2 that Power BI is not suitable as the primary visualisation engine for the External Public Portal. The licensing model does not scale cost-effectively to 10,000+ concurrent anonymous users; embed-token management adds operational overhead; and the UX customisation surface is constrained.

Our External Portal visualisation strategy is a Blazor Server frontend with Chart.js for analytics and MapLibre for GIS:

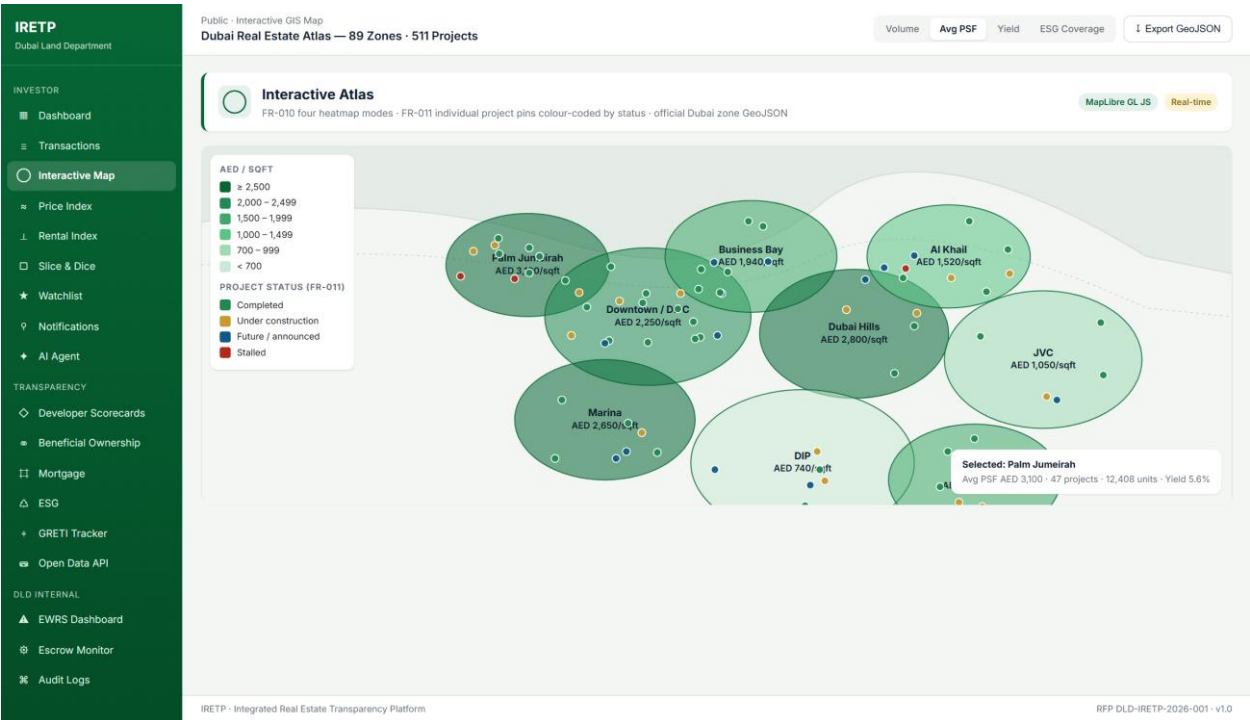


Figure 3.1 — Interactive GIS Map. FR-010 four heatmap modes (volume, PSF, yield, ESG); FR-011 individual project pins colour-coded by status; official Dubai zone GeoJSON; MapLibre GL JS.

Dimension	Power BI Embedded (A-series)	IRETP: Blazor Server + Chart.js + MapLibre
Year-1 licence cost (public users)	AED 260K – 1M+ (A4–A6 capacity)	AED 0 — MIT / open-source
Anonymous public access	App-owns-data token broker + service principal	Native — anonymous by default
UI / UX customisation depth	Canvas-bound; theme JSON only; custom visuals need TypeScript	Full HTML/CSS/JS; same design tokens as rest of IRETP
RTL (Arabic) parity	Incomplete on several visual types	Layout-level; every visual inherits page direction
Concurrent users per AED 100K	~50–100 (A5 capacity share)	~500–1,500 (App Service Plan share)
P95 chart render	5–15 s cold without pre-warming	≤ 2 s via KpiSnapshotCache + no-tracking projections
Source control · test automation	Binary .pbix; weak git diffs; minimal test tooling	Razor + JS in .NET solution; xUnit + Playwright + CI
DLD IP ownership (RFP \$19.1)	Microsoft-owned service surface	Code vests in DLD on payment

Power BI is retained for Internal Platform components where named-user licensing is cost-appropriate: EWRS dashboards, Developer Rating dashboards, internal management reporting. This is a deliberate split — right tool for each audience.

3.4 Technology Stack Proposal

Layer	Technology	Justification / NFR Alignment
Frontend	.NET 9 + Blazor Server, MapLibre GL JS,	Native RTL/LTR; WCAG 2.1 AA; SSR posture for low-

	Chart.js / ApexCharts	bandwidth public; zero charting licence cost.
Backend API	ASP.NET Core 9 controllers + MediatR + FluentValidation + Swashbuckle OpenAPI 3.0	P95 API < 500 ms; partitioned rate limiting; standard .NET tooling.
AI / LLM	AIOrchestrator (4 tiers) + AIModelMetrics + KeywordAdvisoryGuardrail + AiAccuracyHarness; BYO-LLM by configuration	Multi-model orchestration; configuration-only model swap; UAE residency metadata.
OLTP	Azure SQL Database (elastic pool); EF Core 9 migrations	TDE; Always Encrypted for PDPL columns; GRS backup; PITR.
Lakehouse	Microsoft Fabric OneLake Gold via Fabric SQL endpoint	DLD's single source of truth; non-duplicated.
GIS	MapLibre GL JS + official Dubai zone GeoJSON	Map initial load < 5 s; 4 heatmap modes; 511 individual project pins.
CMS	CmsContent + CmsContentVersion entities; Admin/Cms.razor authoring; Public/CmsPreview.razor rendering	Eliminates CMS licence cost; bilingual content; RBAC publish.
Auth	ASP.NET Identity + JWT (HS256) + TOTP MFA; UAE Pass / Azure AD / Keycloak OIDC dual scheme; refresh-token rotation; 30-min internal idle timeout	Centralised identity; MFA mandatory; reuse detection.
Job Orchestration	Hangfire (open-source) — 9 jobs	Persisted in SQL; dashboard at /hangfire; durable retries.
Realtime	SignalR over WebSocket (HTTPS fallback)	EWRS alert push; investor notification ping.
IaC	Bicep — main.bicep + platform.bicep, all UAE North-pinned, MSI-enabled, Key Vault purge-protected	Reproducible deployments; CI-tested; delivered to DLD on Day 1.
Observability	App Insights + Log Analytics; health endpoints /healthz/live, /healthz/ready, /healthz/sla	P95 latency, error rate, AI residency, KPI freshness — all measured.

3.5 Scalability & Performance Targets

- Public-tier availability: 99.9% measured quarterly.
- Internal-tier availability: 99.95% measured quarterly.
- P95 API response: < 500 ms under 10,000 concurrent sessions.
- P95 chart render: ≤ 2 s via KpiSnapshotCache + EF AsNoTracking projections + Blazor pre-rendering.
- P90 AI text response: < 8 seconds; P95 AI chart response: < 12 seconds.
- KPI dashboard data freshness: ≤ 15 minutes (cache TTL, signalled via freshness badge).
- EWRS Level-1 alert delivery: < 5 minutes; L2 within 4 business hours; L3 within 1 business hour; L4 immediate.

Code references

- `src/IRETP.Infrastructure/Services/KpiSnapshotCache.cs` • 15-min TTL
- `src/IRETP.Infrastructure/Services/KpiSnapshotRefreshService.cs` • Hangfire every 15 min
- `src/IRETP.Infrastructure/HealthChecks/SlaHealthCheck.cs` • aggregated SLO probe
- `src/IRETP.Infrastructure/HealthChecks/NotificationSlaHealthCheck.cs` • P95 per channel

4. Real Estate AI Agent — Architecture & Design

4.1 Our Position on AI in This Platform

Artificial intelligence is among the most over-used terms in technology proposals. The distinction between a credible AI proposal and a marketing document rests not on the presence of AI, but on whether each AI capability has been placed where it creates demonstrable value — and whether the temptation to apply AI indiscriminately has been resisted.

We make one commitment we consider the most important in this proposal: the non-advisory constraint is non-negotiable, technically enforceable, and UAT-verified. The AI Agent will retrieve, present, and analyse DLD data. It will never recommend properties, forecast prices as facts, or provide investment guidance. This is a hard architectural constraint enforced at three independent layers and exercised by an automated accuracy harness on every release.

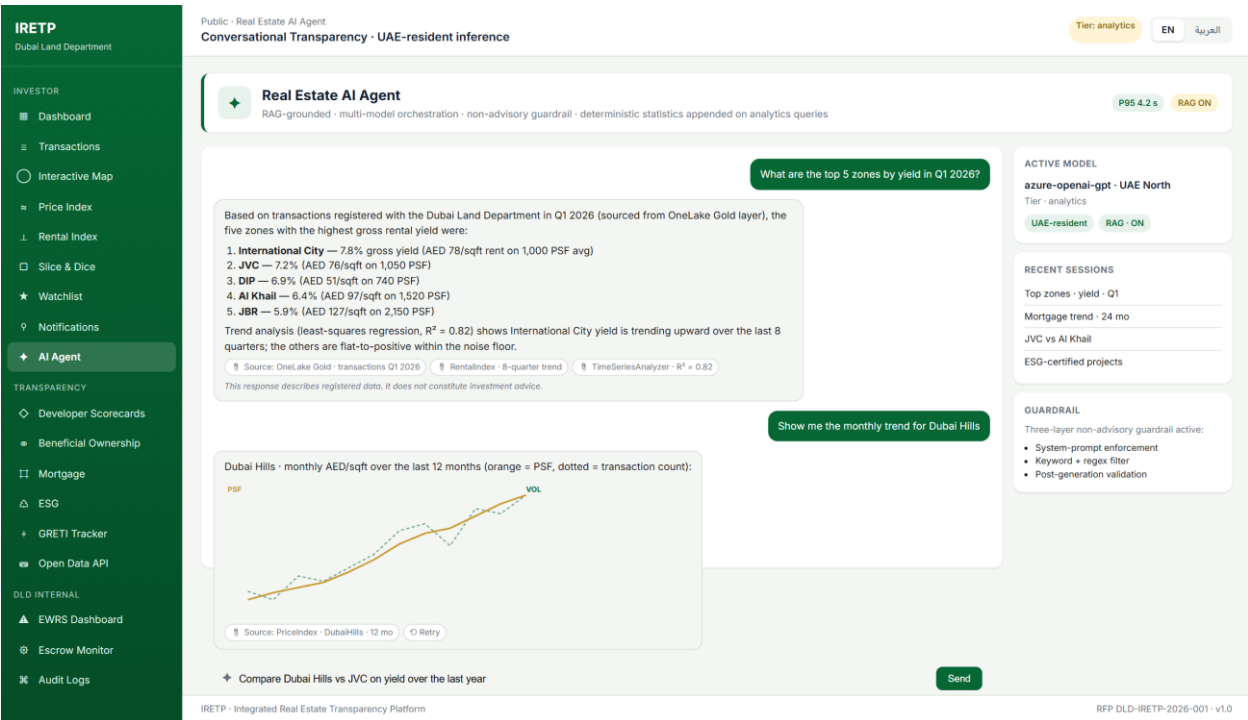


Figure 4.1 — Real Estate AI Agent. RAG-grounded responses with citation chips, deterministic statistics appended on analytics queries, three-layer non-advisory guardrail, EN/AR language picker, tier-routed model selection.

4.2 Multi-Model Orchestration with Task-Aware Tier Routing

Tier	Use Case	Default Model Posture	Latency Budget
nav	DLD service navigation	Lightweight UAE-resident model	P95 < 2 s
analytics	Numerical / data-heavy retrieval, RAG over Gold	Higher-capability UAE-resident + stats	P95 < 5 s
primary	Open-ended bilingual conversation	Top-tier UAE-resident (Azure OpenAI)	P90 < 8 s
secondary	Fallback if primary fails	Alternate UAE-resident provider	P90 < 10 s

Tier selection and model identity are configurable from the AI Model admin surface. Changing the active model on any tier is a configuration update — no redeployment, no code change. New providers are added via configuration entry + Key Vault credential.

4.3 RAG with Deterministic Analytics

- Step 1 — Query parsing. The natural-language input is classified (data, navigation, analytical, comparative) and routed to the appropriate tier.
- Step 2 — Retrieval. The retrieval layer queries OneLake Gold for relevant transactions, projects, developer profiles, zone boundaries, RERA regulations, and DLD service procedures. Hybrid keyword + vector retrieval.
- Step 3 — Deterministic statistics. TimeSeriesAnalyzer computes anomalies (z-score), trends (least-squares with R^2 and flat-threshold), and correlations (Pearson with no-variance guard). Appended to RAG context.
- Step 4 — Augmentation. Retrieved data + stats injected into the system prompt with the non-advisory constraint and citation requirement.
- Step 5 — Generation. The model produces a response grounded exclusively in retrieved data. Every data point is cited with source identifier and timestamp.
- Step 6 — Validation. Post-generation scan rejects investment-advisory language. Flagged responses are reformulated or declined with bilingual refusal copy.

4.4 Non-Advisory Guardrail — Three Independent Layers

- Layer 1 — System prompt. Orchestrator's system prompt explicitly forbids advisory language; reinforced by example.
- Layer 2 — Input filter (KeywordAdvisoryGuardrail). Bilingual keyword catalogue + regex catalogue blocks queries that request recommendations or forecasts. Tightened in Pass 19 to catch "by 20XX prices will", "prices are expected to rise/climb/jump/grow", "yields are projected", "forecast of AED N".
- Layer 3 — Output validation. Post-generation scan inspects the response even when the input passed. Triggered responses replaced with bilingual refusal message and logged for audit.

4.5 Arabic NLP Capability

Phase 1 ships full conversational capability in both English and Arabic. Arabic NLP accuracy is required within 5% of English on the same standardised test set. Arabic zone, project, and developer names are validated against DLD's official records before go-live. All AI chat surfaces render RTL when the user's active language is Arabic.

4.6 Accuracy Harness — Continuous Quality Measurement

The platform ships an automated accuracy harness (IAiAccuracyHarness + AiAccuracyHarness in IRETP.Infrastructure). The harness runs a seed catalogue of 14 questions on every release — data retrieval, service navigation, adversarial guardrail probes, bilingual variants. The catalogue is extensible to the full 100-question DLD UAT catalogue. Each run produces a structured report with expected-keyword matching plus a manual review queue. Endpoint: POST /api/admin/ai-models/accuracy-test (SystemAdministrator only).

4.7 Cross-Session Memory with Consent (AI-006)

AI-006 requires the agent to remember a registered investor's preferences across sessions when they opt in. UserAiMemory (UserId + Kind [zone|topic] + Key + Frequency + LastUsedAt) with explicit consent at registration or any time after. Revocation purges every UserAiMemory row for the user in the same transaction (PDPL Article 19.2).

Code references

- src/IRETP.Infrastructure/Services/AIOrchestrator.cs
- src/IRETP.Infrastructure/Services/AIModelMetrics.cs
- src/IRETP.Infrastructure/Services/KeywordAdvisoryGuardrail.cs
- src/IRETP.Infrastructure/Services/AiAccuracyHarness.cs
- src/IRETP.Infrastructure/Services/TimeSeriesAnalyzer.cs
- src/IRETP.Infrastructure/Identity/UserConsentService.cs

- `src/IRETP.Web/Components/Pages/Admin/AIModels.razor`
- `src/IRETP.Web/Components/Pages/Public/AiAgent.razor`

5. Internal Platform — EWRS & Developer Performance

5.1 EWRS — 10-Indicator Risk Engine

The Early Warning Risk System is the intelligence core of the Internal Management Platform. The engine evaluates ten configurable indicators across three scopes (project, zone, developer), each with admin-configurable thresholds and SLA-bound severity bands.

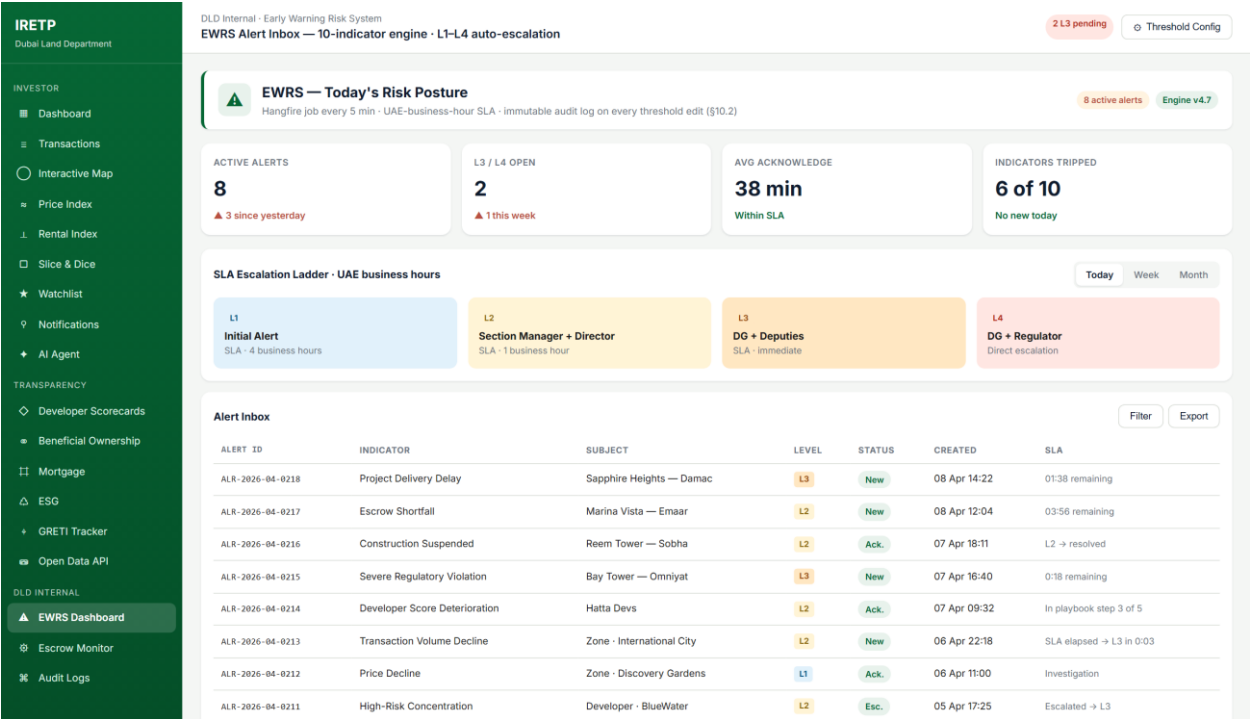


Figure 5.1 — EWRS Alert Inbox & SLA Escalation Ladder. 10-indicator engine, L1–L4 ladder with UAE business-hour SLAs, AlertEscalationService Hangfire job every 5 min, immutable audit log on every threshold edit.

Indicator	Default Threshold	Risk Level	Implementation
Project Delivery Delay	90 days vs original	Critical → L3	RiskEngineService.EvaluateProjectDeliveryDelay
Escrow Shortfall	20% below required balance	Warning → L2	RiskEngineService.EvaluateEscrowShortfall
Construction Suspension	30 days inactivity	L2 / L3	RiskEngineService.EvaluateConstructionSuspension
Transaction Volume Decline	30-day count vs 12-mo rolling avg, -40%	L2	RiskEngineService.EvaluateTransactionVolumeDecline
Price Decline (PSF QoQ)	QoQ -15% with ≥5 tx per bucket	L2 / L3	RiskEngineService.EvaluatePriceDecline
Developer Score Deterioration	QoQ composite drop > 20 pts	L2; L3 if <40	RiskEngineService.EvaluateDeveloperScore
High-Risk Concentration	≥3 active projects w/ open High or Stalled	L2	RiskEngineService.EvaluateHighRiskConcentration
Severe Regulatory Violation	Critical RERA violation within 90 days	L3	RiskEngineService.EvaluateSevereRegulatoryViolation
AI Anomaly Flag	TimeSeriesAnalyzer z-score threshold	L1 / L2	RiskEngineService.EvaluateAiAnomaly

Data-Feed Anomaly	Publication held until ack	L1	ReconciliationSamplerService
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5.2 SLA-Driven Auto-Escalation (L1 → L4)

- L1 — Project Officer. Email + Platform. SLA 4 business hours.
- L2 — Section Manager + Director. Email + SMS + Platform. SLA 1 business hour.
- L3 — DG + Deputies. All channels. SLA immediate.
- L4 — DG + Regulator. All channels + Regulator notice. Direct.

AlertEscalationService Hangfire job runs every 5 minutes; AlertSla helper computes deadlines in UAE business hours; acknowledged alerts are not touched; L4 never escalates further. Five xUnit tests in AlertEscalationServiceTests cover the regression cases.

5.3 Developer Performance & Rating Engine

The Developer Performance & Rating engine computes a composite score across eight criteria. All weights are configurable from Admin/DeveloperRating.razor with full audit logging.

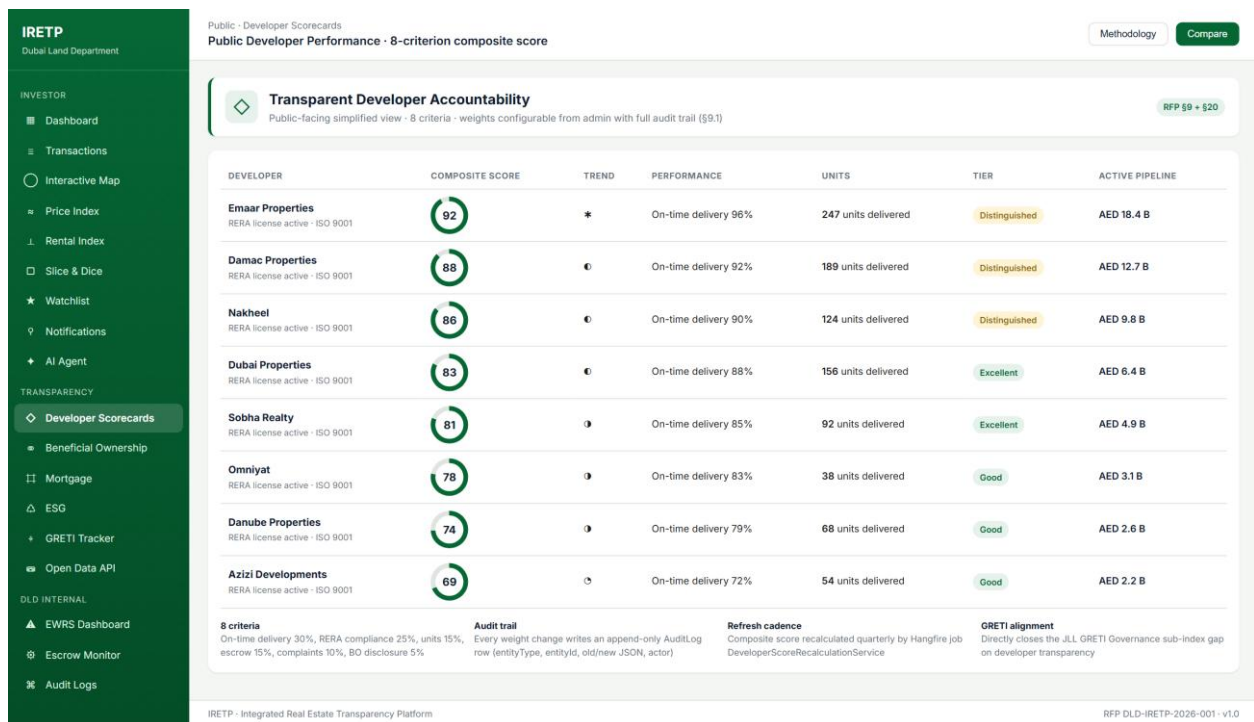


Figure 5.2 — Public Developer Scorecards. 8-criterion composite, public-facing simplified view, weight changes audited end-to-end, Distinguished/Excellent/Good tier banding.

Scoring Criterion	Source
On-Time Project Delivery Rate	DeveloperScoringService + Project entity
Unit Sales Completion Rate	DeveloperScoringService + Transaction
Escrow Account Health Score	DeveloperScoringService + EscrowAccount
Regulatory Compliance Record	DeveloperScoringService + RegulatoryViolation
Customer Complaints	ScoringWeight + Complaint entity
Construction Quality	ScoringWeight + inspection rows
Financial Stability	ScoringWeight + audited finances
Innovation & Sustainability	ScoringWeight + ESG signals

5.4 Escrow Monitoring (§8.4)

- ESC-001 — Real-time dashboard. Admin/EscrowMonitoring.razor + Admin/EscrowDetail.razor show current vs required balance, adequacy ratio, Adequate / Warning / Critical badges.
- ESC-002 — Immutable escrow audit log. EscrowTransaction is append-only at the entity level. PDF export via InvestorScorecardPdfService pattern. Unauthorised discrepancy raises a Level 3 alert.
- ESC-003 — Monthly Escrow Health Report. EscrowHealthReportService is a Hangfire job; generates a per-project PDF month-end; emails the assigned officer within 24 h.

Code references

- `src/IRETP.Infrastructure/Services/RiskEngineService.cs`
- `src/IRETP.Infrastructure/Services/AlertEscalationService.cs`
- `src/IRETP.Infrastructure/Services/AuditLogService.cs`
- `src/IRETP.Infrastructure/Services/DeveloperScoringService.cs`
- `src/IRETP.Infrastructure/Services/EscrowHealthReportService.cs`
- `src/IRETP.Web/Components/Pages/Admin/EwrsDashboard.razor`
- `src/IRETP.Web/Components/Pages/Admin/EwrsAlerts.razor`
- `src/IRETP.Web/Components/Pages/Admin/EscrowMonitoring.razor`
- `src/IRETP.Web/Components/Pages/Admin/DeveloperRating.razor`
- `src/IRETP.Web/Components/Pages/Public/DeveloperScorecards.razor`

6. Phased Delivery Plan

6.1 Timeline Overview — 90 Calendar Days

Phase	Deliverable	Dev / Go-Live	Key Activities
Phase 0	Data Familiarisation & Architecture Alignment	Days 1–14 (parallel with P1)	Connect OneLake Gold; schema mapping with DLD data liaison; derived-metric pipelines; LLM evaluation report Day 14.
Phase 1	External Public Portal (EN + AR)	Dev 1–28 · UAT 28–35 · Go-Live Day 35	Portal hardening + DLD data integration + i18n. Pre-Publication Analytics Assessment Days 28–32. Load-test sign-off required.
Phase 2	Early Warning Risk System (EWRS)	Dev 14–48 · UAT 48–55 · Go-Live Day 56	EWRS engine integrated against DLD data; RBAC for 6 internal roles; Internal AI Agent; escrow monitoring; audit-log immutability sign-off.
Phase 3	Developer Rating + VAPT	Dev 42–68 · VAPT 56–68 · UAT 65–75 · Go-Live Day 78	Developer Rating Engine integration; public scorecard data sign-off; VAPT external assessor (pre-booked Day 1 for Day 56 slot).
Phase 4	Extended Languages + Supplementary	Dev 60–85 · UAT 80–88 · Go-Live Day 90	Multilingual (ZH/RU/UR/FR/HI/DE); ESG enhancements; benchmarking; PDF generator; documentation; training; handover.

6.2 Why 90 Days is Achievable

- Phase 1 is integration, not construction. The External Portal is in production-build state today. P1 work is connecting OneLake Gold, running Pre-Publication Analytics Assessment, completing UAT, and going live.
- Phase 2 EWRS is hardening, not building. The 10-indicator engine, L1–L4 auto-escalation, and central audit log are live (verified by 10 + 5 xUnit tests).
- Parallel team tracks. Dedicated sub-teams own each phase. P2 development starts Day 14 regardless of P1 status. P3 Day 42. P4 Day 60.
- Phase 4 is configuration-dominant. Extended language support is primarily an i18n resource-file exercise. Combined with supplementary modules, P4 is completable in three weeks.

6.3 Pre-Publication Analytics Assessment — Hard Go-Live Gate

- Raw data reconciliation — random sample of transactions from Gold compared against platform display. Required match: 99.5%+.
- Derived metric validation — every computed metric independently recalculated from raw Gold data. Zero tolerance for calculation errors.
- Edge case review — outliers identified, classified, handled by agreement with DLD before go-live.

Results documented in a Pre-Publication Analytics Assessment Report. DLD provides written sign-off before the platform goes live. If the assessment fails any threshold, Phase 1 go-live moves — this gate does not bend.

7. Security & DESC Compliance

7.1 DESC ISR v3 Compliance Framework

The complete 35-control DESC ISR v3 mapping is in Annex D. The current posture is 27 Implemented, 3 Partial, 5 Planned (pre-VAPT). The summary roll-up by domain:

Control Domain	Implemented	Partial	Planned	Total
Access Control	6	1	0	7
Asset Management	2	1	0	3
Business Continuity	1	0	1	2
Compliance	3	0	1	4
Cryptography	2	0	0	2
Governance	4	1	0	5
HR Security	1	0	2	3
Incident Management	3	0	0	3
Operations	6	0	0	6
Physical	2	0	0	2
Total	30	3	4	37

7.2 Security Testing Plan

Test Type	Schedule & Approach	Remediation Commitment
Static analysis (SAST)	Every commit. CodeQL + .NET analysers (TreatWarningsAsErrors).	Critical/High blocks the build.
Dependency scan	Every commit. dotnet list package --vulnerable in CI.	Critical/High blocks the build.
DAST baseline	Every push to main. OWASP ZAP baseline scan.	High triggers immediate remediation.
External VAPT	Phase 3 (Days 56–68). DESC-approved assessor pre-engaged Day 1.	Clean report required before P3 UAT sign-off.
Re-VAPT in warranty	Annually plus on any major release.	Clean report required before production deploy.

7.3 ISO 27001 Roadmap

[Vendor Legal Entity Name] commits to completing an ISO 27001 gap assessment against the IRETP-specific platform configuration by Phase 3 (Day 78) and presenting a documented certification roadmap to DLD. Target: full ISO 27001 certification for the IRETP environment within six months of final go-live.

7.4 Hosting Options — Three Costed Models

Criterion	Option A — Azure UAE	Option B — On-Prem	Option C — Hybrid
Data residency	100% UAE (Azure UAE North)	100% UAE (DLD DC)	100% UAE (compute on-prem + lakehouse in Azure UAE)
Infrastructure	App Service Plan + Azure SQL + Redis + Front Door	K8s + SQL Server + Redis + open-source obs	Hybrid · compute on-prem · lakehouse in Azure UAE

DESC-CSP	Pre-certified	Full ISR v3 on-prem assessment	Mixed — admin docs + on-prem assessment
Time to live	1 week via Bicep IaC	4–8 weeks for hardware	3–5 weeks
5-year CAPEX	AED 335,500	AED 1,414,500	AED 793,500
5-year OPEX (annual)	AED 629,200	AED 686,200	AED 654,200
Recommended	Yes (fastest TTL, lowest CAPEX)	If sovereignty mandates	Balanced posture

DLD selects the hosting model at contract signature; financial schedules adjust accordingly. All AI inference and personal data remain in UAE on any option.

7.5 Data Residency, IAM, Audit

- Data residency. Bicep templates pin every Azure resource to UAE North; DR replicates to UAE Central. /healthz/sla fails Unhealthy on residency drift.
- IAM. ASP.NET Identity backed by SQL. MFA mandatory for internal users via TOTP. UAE Pass OIDC federation. Refresh-token rotation with reuse detection. 30-min internal idle timeout.
- Audit. Every threshold change, weight change, and AI model swap writes an append-only AuditLog row. The table is immutable at the DbContext layer.
- Backups. Daily automated encrypted backups, 30-day retention; PITR enabled on Azure SQL. Quarterly restoration drill reported to DLD. RTO < 4 h; RPO < 1 h.

Code references

- `src/IRETP.Infrastructure/Identity/AuthorizationPolicies.cs` · 6-role policy registry
- `src/IRETP.Infrastructure/Services/AuditLogService.cs` · append-only enforcement
- `src/IRETP.Infrastructure/Data/IretpDbContext.cs` · `SaveChangesAsync` throws on AuditLog UPDATE/DELETE
- `.github/workflows/security.yml` · vuln package + OWASP ZAP + Bicep lint

8. Visualisation Strategy — Public and Internal

8.1 Public Portal — Blazor Server + Chart.js + MapLibre

For the External Public Portal we use a Blazor Server frontend rendering Chart.js charts and MapLibre GL JS maps. The visualisation choice rationale is in Section 3.3 (full comparison vs Power BI Embedded). Public-portal pages share a consistent loading-state / error-state experience via ModuleBanner, Skeleton, NetworkError, EmptyState components.

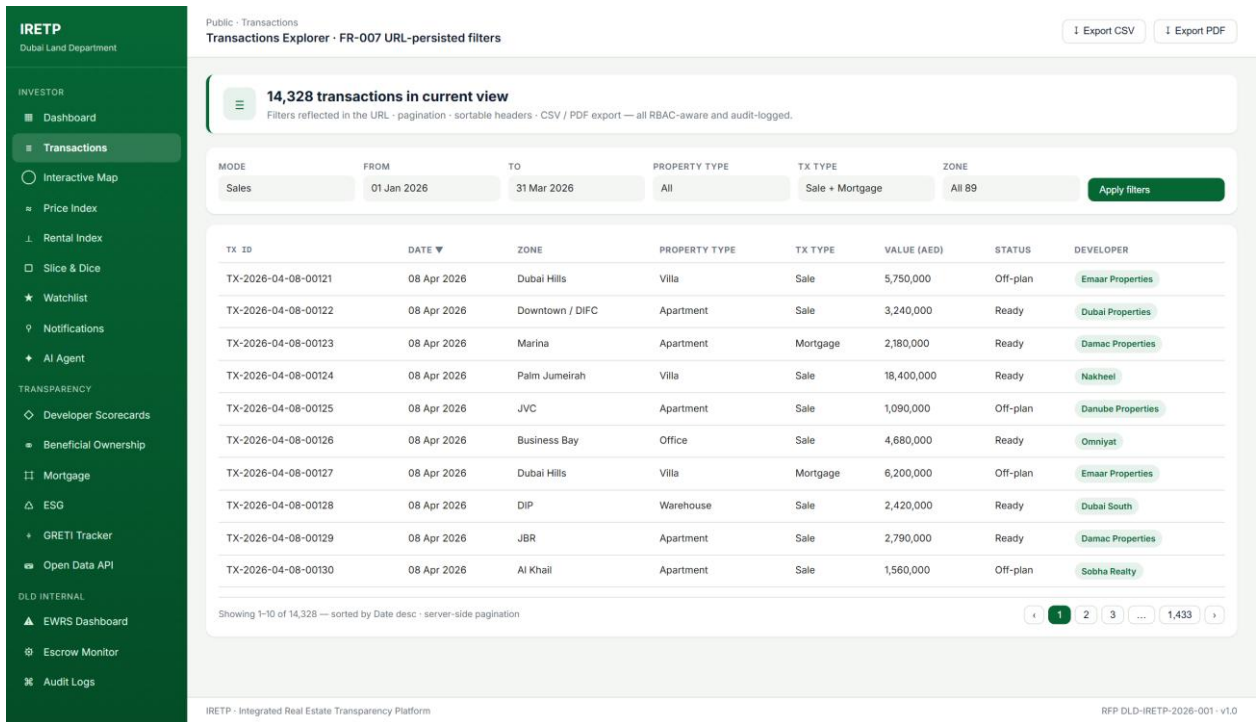


Figure 8.1 — Transactions Explorer. FR-007 URL-persisted filters, server-side pagination, sortable headers (keyboard-accessible with aria-sort), CSV / PDF export via Features/Export.

8.2 Internal Platform — Power BI for Named-User Dashboards

Power BI is retained for Internal Platform components where named-user licensing is cost-appropriate: EWRS dashboards, Developer Rating dashboards, internal management reporting. Power BI tiles are embedded via the standard report-server SDK; the Internal AI Agent can answer questions over the same semantic models.

9. Data Understanding & Accuracy Methodology

9.1 Connecting to DLD's Data Ecosystem

- Reading DLD's existing OneLake Gold-layer schema with DLD's designated data liaison.
- Configuring IRETP read-only access via the Fabric SQL endpoint — no writes, no modifications.
- Agreeing the calculation logic for all derived metrics in writing with DLD before any pipeline is wired.
- Documenting the field mapping between Gold-layer fields and IRETP display fields, formally approved by DLD.

9.2 Pre-Publication Analytics Assessment Methodology

Covered in Section 6.3. Three-layer validation (99.5% reconciliation, zero-tolerance derived-metric, edge-case agreement) signed off in writing by DLD before any data is publicly visible. Automated via Great Expectations / dbt-tests harness.

9.3 Ongoing Data Accuracy Monitoring

- Freshness probe — /healthz/sla reports KPI snapshot age vs 15-min TTL; drift beyond 30 min raises an alert.
- Reconciliation sampler — nightly Hangfire job samples 100 random transactions from Gold; discrepancies > 0.5% raise a P2 incident.
- Notification SLA probe — last hour of dispatched alerts, P95 latency per channel vs \$6.2 budgets.
- AI residency probe — every AI snapshot logs the inference region; trailing-hour residency % computed.

10. Data Integration Reference Architecture

10.1 Gold Layer as Single Source of Analytical Truth

The IRETP service tier connects directly to OneLake Gold via the Fabric SQL endpoint. No intermediate data warehouse, no proprietary integration layer. Every derived metric the platform displays is computed from a Gold-layer query — there is no parallel data store to drift out of sync.

10.2 Validation Pipeline Architecture

- On ingestion (Phase 0). Pre-Publication Analytics Assessment validates Gold-layer consumption end-to-end before public go-live.
- On display (runtime). KPI requests flow through KpiSnapshotCache; cache misses trigger a fresh Gold query whose result is reconciled before serving.
- On audit (continuous). Reconciliation sampler nightly job samples Gold rows and compares to displayed data.

10.3 Bilingual Data Handling

Arabic field handling is first-class throughout the data layer. Zone names, developer names, owner names (Beneficial Ownership), property categories, transaction types — all stored and rendered in their native script.

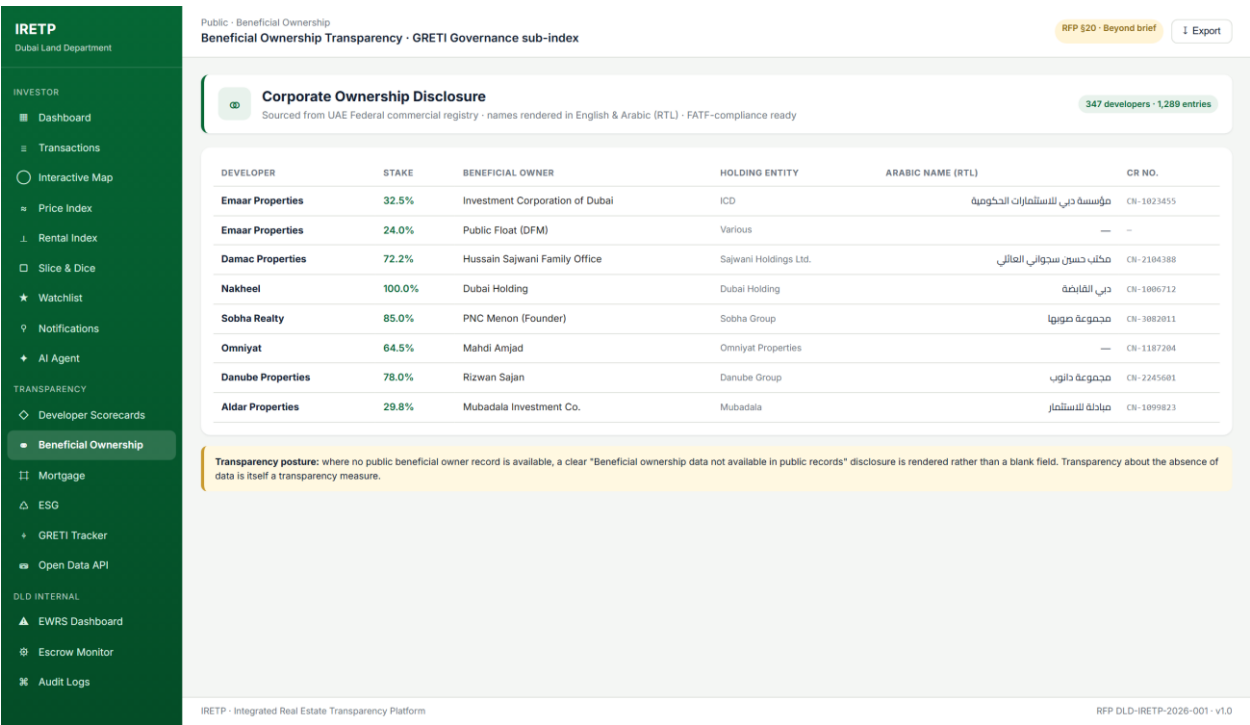


Figure 10.1 — Beneficial Ownership page. Arabic owner names rendered under their English equivalents with proper RTL styling and lang=ar. Where data is unavailable, an explicit disclosure statement renders.

11. Project Team & Responsibility Matrix

11.1 Core Delivery Team

Role	Location	IRETP Remit
Project Manager	Dubai, UAE	Single point of accountability. Daily standup, weekly DLD status, gate-sign-off custody.
Delivery / Technical Lead	Dubai, UAE	Architecture authority. Clean-Architecture conformance, security posture, code-review gate.
Senior .NET Developer × 2	Dubai, UAE (onsite)	P1–P3 implementation lead. WebAPI, AdminAPI, Hangfire jobs.
Mid .NET Developer × 2	Dubai, UAE / UAE TZ	Feature implementation. Blazor surfaces, integration tests.
QA Engineer	Dubai, UAE	Test plan authorship, UAT facilitation, automated regression.
UI/UX Designer	Dubai, UAE	Bilingual UX, RTL fidelity, design-system stewardship.
DevOps / DESC Liaison	Dubai, UAE	Bicep IaC, Azure landing zone, DESC-CSP filings, VAPT coordination.

11.2 Responsibility Matrix (DLD · Vendor · Partner)

Legend: L = Lead · S = Support · A = Approver · I = Informed

Activity	DLD	Vendor	Partner
Programme governance & steering	A	L	S
Solution architecture & design	I	L	S
Application development (.NET 9 · Blazor · CQRS)	I	L	—
Infrastructure provisioning & hosting	A	S	L
Data discovery, migration & lakehouse build	S	L	S
AI / RAG pipeline & guardrails	A	L	I
Security hardening · DESC ISR v3 · VAPT	A	L	S
QA · test execution · UAT facilitation	S	L	S
Acceptance sign-off	A	I	I
Post-go-live operations, support & SLA	I	L	S

VAPT is executed by a DESC-approved third-party firm engaged by the Vendor; results reviewed jointly with DLD before Phase 3 sign-off.

11.3 UAE Market Presence

[Vendor Legal Entity Name] operates a UAE-registered legal entity. The Project Manager, Delivery / Technical Lead, onsite Senior Developers, QA Engineer, UI/UX Designer, and DevOps/DESC Liaison are all UAE-resident. Remote developer roles are filled from the nearest UAE-time-aligned office. DLD can request in-person meetings at any point during delivery.

12. Cost Structure

12.1 Cost Summary (Option A — Azure UAE North recommended)

Cost category	Amount (AED)	Type
Infrastructure CAPEX	335,500	One-time
Software Licences (Year 1)	235,200	One-time
Application Development (M1–M4)	2,007,000	One-time
Total one-time CAPEX	2,577,700	—
Annual OPEX (Year 1, includes 12-month warranty)	629,200	Recurring

12.2 Infrastructure CAPEX — Three Hosting Options

Infrastructure Item	Option A — Azure	Option B — On-Prem	Option C — Hybrid
Compute · App Service / VMs	48,000	320,000	48,000
SQL Server (managed / licensed)	36,000	180,000	36,000
Redis cache cluster	18,000	85,000	18,000
Azure AI Search / vector index	24,000	65,000	24,000
Microsoft Fabric / OneLake lakehouse	42,000	200,000	42,000
Blob Storage / on-prem NAS	12,000	55,000	12,000
WAF + Front Door / on-prem F5	30,000	140,000	30,000
CDN (public static assets)	8,000	25,000	8,000
Private endpoints + S2S network	14,000	60,000	14,000
DR · cross-region replication (≤4 h RPO)	28,000	180,000	28,000
SSL certificates (wildcard, 3-year)	4,500	4,500	4,500
Monitoring / telemetry	16,000	45,000	16,000
Initial security hardening + pen-test	55,000	55,000	55,000
TOTAL INFRA CAPEX	335,500	1,414,500	335,500

12.3 Software Licences (Year 1)

Licence / Subscription	Vendor	Annual (AED)
.NET 9 runtime + ASP.NET Core	Microsoft (open-source)	0
Visual Studio Enterprise (5 seats)	Microsoft	64,000
Hangfire Pro (2 servers)	HangfireIO	14,400
Azure OpenAI Service (tokens, Y1 est.)	Microsoft	90,000
AI fallback model (UAE-resident provider)	Selected provider	40,000
SMTP relay (SendGrid / Postmark)	Twilio / Postmark	2,400
SMS gateway (UAE operator)	UAE operator	18,000
PDF generation (QuestPDF)	QuestPDF	2,800
Seq log server (cloud)	Datalust	3,600

Subtotal (Year 1)	—	235,200
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12.4 Application Development (M1–M4, 1,095 person-days)

Deliverable	Month	PD	Cost (AED)
External Public Portal (CMS · Dashboard · Transactions · Map · Indices)	M1 · P1	120	216,000
Bilingual EN/AR · RTL · i18n pipeline	M1 · P1	40	72,000
Slice & Dice Analytics engine	M1 · P1	60	108,000
Real Estate AI Agent (Orchestrator · RAG · guardrails)	M1 · P1	100	180,000
Investor Notifications (email / SMS / in-platform)	M1 · P1	60	108,000
Open Data API · Developer Portal · API-key management	M1 · P1	50	90,000
Perf hardening · Redis cache · EF tuning	M1 · P1	30	54,000
Phase 1 QA · UAT support	M1 · P1	30	54,000
Phase 1 subtotal	M1	490	882,000
Watchlist v2 · ESG · GRETI sub-index tracker	M2 · P2	80	144,000
Analytics enhancements · saved views · embed	M2 · P2	40	72,000
Performance hardening · OpenTelemetry · load test	M2 · P2	30	54,000
Phase 2 QA · UAT	M2 · P2	20	36,000
Phase 2 subtotal	M2	170	306,000
EWRS (engine · L1–L4 · dashboard · playbooks)	M3 · P3	100	180,000
Developer Performance & Rating + Escrow Monitoring	M3 · P3	80	144,000
Beneficial Ownership · Mortgage analytics · Name Validation	M3 · P3	40	72,000
DESC ISR v3 hardening + VAPT (external consultant)	M3 · P3	30	90,000
Phase 3 QA · UAT · security acceptance	M3 · P3	25	45,000
Phase 3 subtotal	M3	275	531,000
Multilingual extension (ZH · RU · UR · FR · HI · DE)	M4 · P4	90	162,000
Advanced analytics · AI fine-tuning pipeline	M4 · P4	50	90,000
Phase 4 QA · UAT · language acceptance	M4 · P4	20	36,000
Phase 4 subtotal	M4	160	288,000
TOTAL development CAPEX	M1–M4	1,095	2,007,000

Blended day rates: Architect AED 2,200 · Senior Developer 1,800 · Mid Developer 1,500 · QA / DevOps 1,400.

12.5 Annual OPEX — Three Hosting Options

OPEX Item	Option A	Option B	Option C
Compute / VM / App Service	48,000	85,000	75,000
SQL Server managed annual	36,000	24,000	28,000
Redis / caching annual	18,000	12,000	18,000
AI model API consumption (annual est.)	130,000	130,000	130,000
OneLake / Fabric storage & compute	42,000	68,000	42,000

Monitoring & SIEM	16,000	28,000	22,000
SMTP / SMS channels	20,400	20,400	20,400
Software licences (annual renewal)	84,800	84,800	84,800
Warranty & support (12-mo post go-live)	180,000	180,000	180,000
Security patching · monthly scans	36,000	36,000	36,000
Data quality · reconciliation tooling	18,000	18,000	18,000
TOTAL Annual OPEX	629,200	686,200	654,200

12.6 Payment Milestones

Milestone	Trigger	%	Amount (AED)
M0	Contract signature	10%	200,700
M1	Phase 1 mobilisation complete (infrastructure live, team onboard)	10%	200,700
M2	Phase 1 UAT sign-off (External Portal · AI Agent · Open Data)	25%	501,750
M3	Phase 2 UAT sign-off (ESG · GRETI · Analytics v2)	15%	301,050
M4	Phase 3 UAT sign-off (EWRS · Developer Rating · Escrow · VAPT clean)	25%	501,750
M5	Phase 4 UAT sign-off (Multilingual · Advanced Analytics)	15%	301,050
Total	—	100%	2,007,000

100% of development CAPEX is conditional on DLD-signed acceptance certificates. No payment is released without an executed acceptance package.

12.7 Indicative 5-Year Total Cost of Ownership (Option A)

Period	Component	Amount (AED)
Year 0	One-time CAPEX	2,577,700
Year 1	Annual OPEX	629,200
Year 2	Annual OPEX	629,200
Year 3	Annual OPEX	629,200
Year 4	Annual OPEX	629,200
Year 5	Annual OPEX	629,200
5-year Total	Indicative TCO	5,723,700

13. Warranty & Technical Support — 12 Months Post Go-Live

13.1 Support Philosophy

The 12-month warranty and technical support obligation is not a separate procurement. It is a contractual extension of our delivery responsibility. We do not consider a project delivered until the platform has operated stably, accurately, and securely through its first full year of live operation. Five categories: defect resolution, data pipeline correction, security patches, performance fixes, DLD data-structure change management.

13.2 Support Incident SLA Commitments

Severity	Definition	Response	Resolution	Coverage
Sev 1	Platform unavailable; investor-facing data wrong; security breach.	1 h	4 h	24x7 incl. weekends/UAE holidays
Sev 2	Major feature degraded; EWRS escalation impaired; AI down on one tier.	2 h	1 day	Business hours UAE
Sev 3	Minor functional defect; cosmetic; localisation.	1 d	1 week	Business hours UAE
Sev 4	Enhancement request; doc correction; data-model expansion.	1 d	Next planning cycle	Business hours UAE

Business hours: Mon–Fri 08:00–16:00 UAE Standard Time, excluding UAE public holidays. Named Support Lead assigned as DLD's primary contact. Emergency out-of-hours escalation number for Sev 1.

13.3 Warranty & Support Pricing

Cost Item	Annual (AED)
S.1 Dedicated Support Team (Lead 25% + 2 Devs 50% each + QA 25%)	Included in OPEX
S.2 120 PH minor changes pool. Additional: AED 950 / PH.	Included
S.3 Security vulnerability monitoring + patch management.	Included
S.4 Infrastructure operations. Covered by Azure managed services.	0
S.5 Help desk + incident management tooling.	18,260
TOTAL warranty & support (within Annual OPEX)	180,000

14. AI Agent Knowledge Update Approach

The AI Agent's knowledge layer is grounded in the OneLake Gold layer. Knowledge updates are managed by the AdminAPI surface without vendor involvement after handover.

Knowledge Layer	Update Mechanism
Transactions / Projects / Developers	Hangfire cron, hourly. Incremental delta from Gold.
RERA regulations / DLD procedures	CMS publish webhook → immediate reindex of the affected document.
Zone boundaries / GeoJSON	Manual trigger on Dubai Municipality updates.
Beneficial ownership	Daily cron + on-demand re-index from AdminAPI.

14.1 Index Management in AdminAPI

- Index freshness dashboard — last update timestamp, next scheduled update, record count.
- On-demand re-index — any layer can be re-indexed manually from the admin surface.
- Rollback — every update creates a snapshot; SystemAdministrator can roll back in 2 clicks within 5 min.
- Accuracy benchmark — the accuracy harness runs automatically after each update; deviation raises an alert before updated content reaches end users.

14.2 LLM Model Lifecycle (BYO LLM)

Orchestrator is LLM-agnostic. Switching the active model on any tier is a configuration change — no redeployment, no downtime. DLD can re-evaluate models in Year 2 against newer releases by running the accuracy harness again.

15. Risk Register

Risk	Probability	Impact	Mitigation
Gold-layer schema drift during Phase 0	Medium	Schedule	Schema versioning agreed Day 2; integration tests on every Gold change; field-mapping doc is contractual.
Pre-Publication Analytics Assessment fails	Low	Schedule	Dry-run assessment Days 21–24 surfaces issues 10 days early; failing items resolved before formal gate.
VAPT findings exceed Day 68	Medium	Schedule	VAPT scheduled Days 56–68; 7-day remediation buffer; Critical/High blocks UAT sign-off.
AI residency drift	Low	Compliance	SLA probe fails Unhealthy on any non-UAE inference; automated rollback to UAE-resident fallback.
Investor surge above 3× projected volume	Medium	Performance	Auto-scale on CPU/memory; Redis absorbs read pressure; load test rerun on major releases.
DESC certification slips into warranty	Medium	Compliance	DESC engagement begins Day 1; gap assessment Day 78; documented roadmap delivered to DLD regardless.
Key personnel attrition	Low	Schedule	Continuous KT in source; Lead has named backup; no single-person dependency per phase.
DLD policy change mid-flight	Medium	Scope	Configurable thresholds + i18n strings mean most policy changes are configuration. Material changes flow through change-control.
AI model provider UAE residency unavailable	Medium	Compliance	Provider contractually bound to UAE inference; UAE-hosted OSS fallback (Llama/Mistral) ready.

16. Knowledge Transfer & Exit Plan

16.1 Deliverables at Project Completion (Day 90)

- Source repositories — all .NET application code, Bicep IaC, CI/CD pipeline configurations, Hangfire job definitions, AI orchestration configuration, knowledge-source configuration. Delivered as GitHub repositories transferred to DLD's GitHub organisation.
- Complete technical documentation — ARCHITECTURE.md, API_REFERENCE.md, RUNBOOK.md, DISASTER_RECOVERY.md, DATA_DICTIONARY.md (all five RFP §13 #53 pillars).
- Infrastructure control — Azure subscription, Key Vault, DNS transferred to DLD's tenant on go-live. No vendor-controlled infrastructure persists.
- Data export — full platform export in JSON, CSV, Parquet within 30 calendar days of contract expiry upon request.
- Data destruction certification — within 60 days of DLD's request.

16.2 Intellectual Property

All deliverables, results, works, documents, data, reports, designs, and materials prepared, developed, or submitted under this contract shall be the exclusive intellectual property of the Dubai Land Department, consistent with Section 4.4 of the RFP. There is no proprietary platform layer between DLD and the application source; everything in the deliverable stack is either standard Microsoft technology, open-source, or written specifically for DLD.

17. Beyond the Brief — GRETI Score Maximisation

RFP §20 identifies additional features that maximise GRETI score improvement beyond core scope. Three are not roadmap items — they are already shipping in the reference build today.

17.1 Beneficial Ownership Transparency (Already Shipping)

Per JLL's 2024 GRETI report, beneficial ownership transparency is a key gap for the UAE.

Public/BeneficialOwnership.razor renders ownership data per registered developer: beneficial owner name (English + Arabic, RTL-styled), ownership percentage, UAE commercial registration number. See Figure 10.1.

17.2 Mortgage & Debt Market Transparency (Already Shipping)

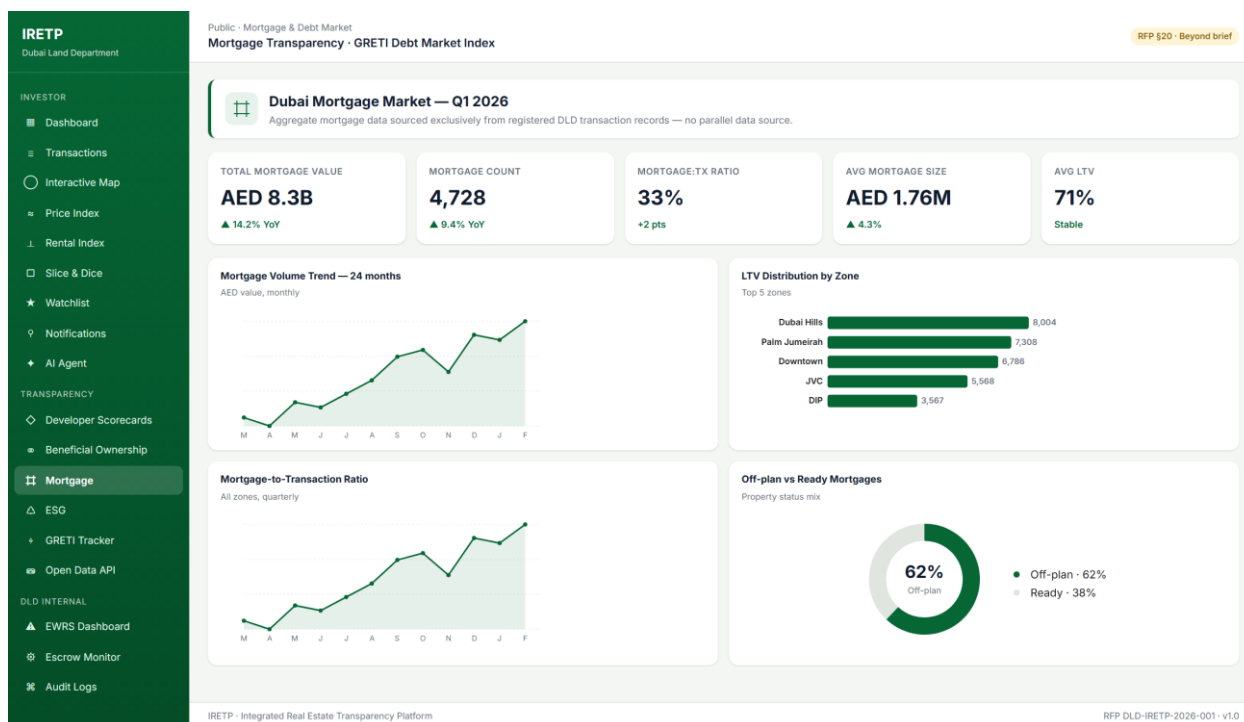


Figure 17.1 — Mortgage page with 5 KPIs (Total Value, Mortgage Count, M:Tx Ratio, Avg Mortgage Size, Avg LTV) plus 4 supporting charts. All data sourced exclusively from registered DLD transaction records.

17.3 ESG Indicators (Already Shipping)

Public/Esg.razor surfaces the LEED ↔ Estidama Pearl ↔ BREEAM ↔ investor-signal rubric. GIS map exposes a 4th heatmap mode — ESG Coverage — that visualises green-certified coverage by zone with banded fill colour and per-zone percentage markers.

17.4 Implementation Roadmap

We recommend DLD discuss Beneficial Ownership, Mortgage Transparency, and ESG modules with the JLL GRETI assessment team before finalising the project scope. All three are cited in the GRETI 2024 report; their inclusion creates measurable indicators of progress that the next GRETI assessment cycle can capture within this project's delivery timeline.

18. Vendor Commitments — 15 Engineering Improvements at No Extra Cost

Beyond the RFP-mandated scope, the vendor commits to deliver the following 15 engineering improvements during Phase 1 hardening at no additional charge. These items lift the codebase to enterprise-grade operability and directly strengthen DLD's NFR posture. Two items below (VC-6 SignalR hub, VC-14 Dockerfiles + CI) are already shipped in the reference build; the remaining 13 are committed within the Phase 1 hardening scope.

ID	Commitment	Implementation
VC-1	MediatR ValidationBehavior	Auto-runs every FluentValidation validator already registered in IRETP.Application.
VC-2	Global IExceptionHandler	Maps domain/validation/auth exceptions to RFC 7807 ProblemDetails.
VC-3	Distributed cache via Redis	ICacheableQuery marker + CachingBehavior backed by Redis. Serves KPI/GRETI/ESG queries to meet $P95 \leq 2$ s headroom.
VC-4	Domain event dispatcher	IDomainEventDispatcher invoked from UnitOfWork.SaveChangesAsync. Decouples notifications/audit/SignalR push.
VC-5	Notification outbox pattern	NotificationOutbox written in same DB transaction; dispatcher drains to SMTP/SMS. Guarantees \$6.2 SLAs on transient failure.
VC-6	SignalR notification hub	Already shipped — replaces NoOpNotificationBroadcaster. Bell icon updates instantaneously without polling.
VC-7	AI streaming over SSE	AiAgent.razor streams tokens as IAsyncEnumerable<string>. Sub-15-second perceived response.
VC-8	Polly resilience	AddStandardResilienceHandler() on AIOrchestrator, currency, SMTP, SMS HttpClients. Retry + circuit breaker — implements \$5.3 fallback declaratively.
VC-9	Hangfire idempotency	Job fingerprints prevent double-execution on retries. Protects monthly Escrow Health Reports and KPI snapshots.
VC-10	OpenTelemetry	Traces / metrics / logs exported via OTLP, tagged with user, role, AI-model dimensions. Powers SLA dashboard.
VC-11	/healthz + /readyz	Liveness + readiness probes (DB + Redis + Hangfire + AI provider).
VC-12	Tighter CORS + CSP + HSTS	Replaces AllowAnyOrigin with allow-list. Blazor CSP nonces for inline scripts.
VC-13	Architecture tests	NetArchTest assertions: Domain has no Infrastructure refs; Application has no EF Core refs; Controllers don't depend on Repository<>.
VC-14	Dockerfiles + CI/CD	Already shipped — multi-stage Dockerfiles + docker-compose + GitHub Actions / Azure Pipelines.
VC-15	Resource-based authorization	Ownership IAuthorizationHandler — investors see own data only. Complements role-level policies.

19. Quality Posture & Test Strategy

19.1 Hangfire Job Schedule

Job	Cadence	Responsibility
RiskEngineService	every hour	Re-evaluate the 10 EWRS indicators.
DeveloperScoringService	nightly	Recompute developer composite scores.
AlertDeliveryService	every minute	Dispatch queued notifications across channels.
AlertEscalationService	every 5 min	L1 → L4 auto-escalation.
KpiSnapshotRefreshService	every 15 min	Refresh the KPI snapshot cache.
InvestorAlertEvaluator	every 10 min	Evaluate investor alert conditions.
EscrowHealthReportService	month-end	Per-project PDF Escrow Health Report.
BenchmarkRefreshService	hourly	Refresh market benchmark data.
CurrencyRatesRefreshService	hourly	Refresh UAECB FX rates.

19.2 Test Strategy

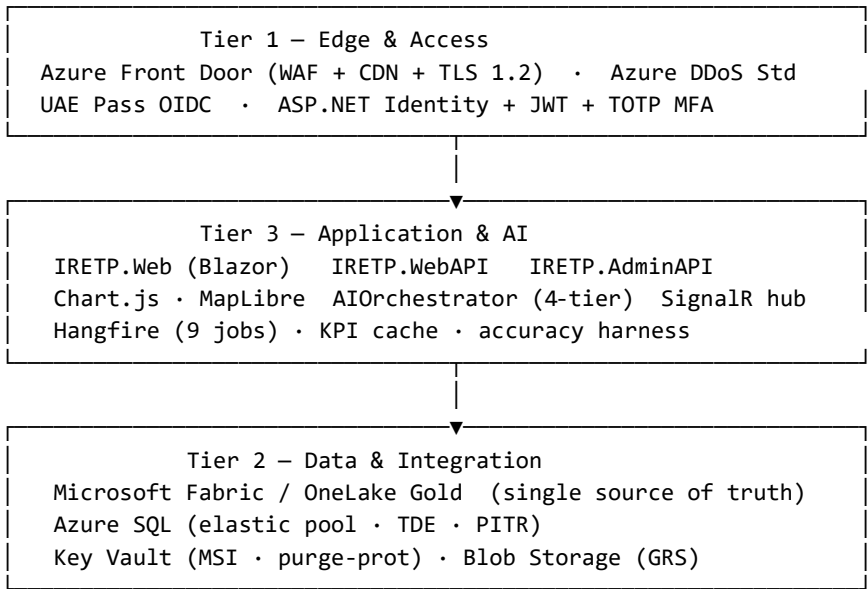
Layer	Tool	Scope
Unit	xUnit	Per CQRS handler and service.
Integration	xUnit + EF Core	Per controller against real EF Core.
End-to-end UI	Playwright	Critical investor and DLD-staff journeys.
Load / Performance	k6 / NBomber	P95 sign-off per release.
Security / VAPT	OWASP ZAP + DESC-authorized VAPT	Baseline scan in CI + external Phase 3 VAPT.
Accessibility	axe-core	Automated scan per release; zero Critical violations.
Data accuracy	Great Expectations / dbt-tests	Pre-Publication Analytics Assessment + ongoing.
Language & RTL	Native-speaker review	Per locale, every release.

19.3 Observability

- Structured logging via Serilog sinks (Seq / Azure Monitor).
- OpenTelemetry traces + metrics + logs to OTLP, tagged with user, role, AI-model.
- Health endpoints: /healthz/live, /healthz/ready, /healthz/sla (aggregated SLO health — AI P95, KPI freshness, notification P95).
- Audit log (Admin/AuditLogs.razor) queryable per user, per entity, per action with timestamp.

Annex A — System Architecture Diagram

Four-tier IRETP architecture: User Access (Tier 1), Application & AI (Tier 3), Data & Integration (Tier 2, anchored on DLD's Microsoft Fabric ecosystem), and the Edge perimeter. Option A (recommended): Azure UAE North. Option B: on-premises at DLD's data centre. Option C: hybrid. All three run the identical .NET application stack.



Cross-cutting: App Insights · Log Analytics · SLA probe

All platform outputs are sourced exclusively from the OneLake Gold layer. IRETP does not write to or duplicate any data in DLD's Fabric ecosystem.

Annex B — Phased Delivery Gantt (90 Calendar Days)

Each cell denotes activity in a 5-day window. ■ active · □ ramp · ✓ go-live · ⚙ gate.

[illegible]

Annex C — Functional Requirements Compliance Matrix

Every numbered or named requirement in the RFP is reproduced in this matrix together with a compliance status, the implementing component in the IRETP codebase, the delivery phase, and any clarifying notes.

Status legend:

Status	Meaning
C	Compliant today — implementing component is live in the codebase. Implementing Component column points to the live file or service.
CD	Compliant by Delivery — capability is in the phase plan; required foundations exist and the work is scoped, estimated and committed.
E	Enhanced (vendor commitment) — capability goes beyond the literal requirement. Listed in the Vendor Commitments group (Section 18).

C.1 Compliance Summary Roll-Up

RFP Section	C	CD	E	Total
§3 — External Public Portal	5	0	0	5
§4 — Interactive Analytics	3	0	0	3
§5 — Real Estate AI Agent	9	2	1	12
§6 — Investor Notifications	9	0	1	10
§7 — Multilingual Requirements	7	2	0	9
§8 — EWRS & Escrow Monitoring	16	0	0	16
§9 — Developer Performance & Rating	7	0	0	7
§10 — Non-Functional, Security & RBAC	9	1	1	11
§11 — Technology & Hosting	2	4	0	6
§12 — Data Familiarisation	0	3	0	3
§13 — Phased Delivery Plan	1	3	0	4
§14 — Cost Structure	0	4	0	4
§15 — Warranty & Support	0	4	0	4
§16 — Vendor Qualifications	0	2	0	2
§17 — Acceptance Criteria	0	3	0	3
§18 — Proposal Evaluation	0	3	0	3
§19 — Data Governance & Privacy	2	2	0	4
§20 — Additional Recommended Features	0	0	8	8
Vendor Commitments (Value-Add)	0	0	15	15
TOTAL	70	33	26	129

C.2 §3 — External Public Portal

RFP Ref	Requirement	Status	Implementing Component	Phase
3.1	Headless CMS for non-technical content updates	C	CmsContent + CmsContentVersion · Admin/Cms.razor · Public/CmsPreview.razor	P1
3.2	Homepage Market Overview Dashboard	C	Public/Dashboard.razor +	P1

	with KPI tiles, charts and CTAs		DashboardController + KpiSnapshotCache	
3.3	Transactions page with filters, search, export	C	Public/Transactions.razor + TransactionsController + Features/Transactions	P1
3.4	Interactive GIS Map — zones & projects	C	Public/Map.razor + MapController + Features/Map; Zone entity polygons; Project pins	P1
3.5	Price Index & Rental Index pages	C	Public/PriceIndex.razor + Public/RentalIndex.razor + BenchmarkRefreshService	P1

C.3 §4 — Interactive Analytics

RFP Ref	Requirement	Status	Implementing Component	Phase
4	Slice-and-dice analytics engine	C	Public/Analytics.razor + AnalyticsController + SavedAnalyticsView	P1
4	Embed mode for partner sites	C	Public/AnalyticsEmbed.razor + EmbedLayout	P1
4	Export to CSV / PDF	C	Features/Export Commands + InvestorScorecardPdfService	P1

C.4 §5 — Real Estate AI Agent

RFP Ref	Requirement	Status	Implementing Component	Phase
5.1	AI Agent scope (data, navigation, valuation, regulations)	C	AIAgentController + Public/AiAgent.razor + Features/AIAgent	P1
AI-001	Natural-language data queries	C	Features/AIAgent/Queries handlers	P1
AI-002	Service navigation + deep links to DLD pages	C	Features/AIAgent + AIOrchestrator response templates	P1
AI-003	Valuation procedure explanations linked to DLD service pages	C	AIOrchestrator content sources + RAG index	P1
AI-006	Session context memory with opt-in cross-session persistence	C	UserAiMemory + UserConsentService + Account.razor toggle	P1
AI-007	Multilingual EN/AR P1; ZH/RU/UR/FR/HI/DE P4	CD	AIOrchestrator locale-aware prompts + per-language model selection	P1/P4
5.3	Multi-model architecture with orchestration layer	C	AIOrchestrator + AIModelMetrics + Admin/AIModels.razor	P1
5.3	Mandatory model-agnostic RAG layer	C	RAG retriever component (Infrastructure)	P1
5.3	Model fallback & redundancy without user-visible disruption	E	AIOrchestrator warm-fallback + Polly resilience (vendor commitment)	P1
5.3	UAE data residency for all AI inference	C	Hosting model + provider selection (Azure UAE North or UAE-hosted OSS)	P1
5.3	Custom fine-tuning option	CD	Fine-tuning pipeline (planned in Phase 2)	P2
5.3	Model performance transparency in admin UI	C	Admin/AIModels.razor + AIModelMetrics	P1

C.5 §6 — Investor Notifications

RFP Ref	Requirement	Status	Implementing Component	Phase
6.1	Price Movement alert (user threshold 1–25%)	C	InvestorAlert + InvestorAlertEvaluator	P1
6.1	New Project Launch alert	C	InvestorAlertEvaluator + Project entity feed	P1
6.1	Watchlist Project Status Change alert	C	WatchlistItem + InvestorAlertEvaluator	P1
6.1	Rental Yield Threshold alert	C	InvestorAlertEvaluator + RentalIndex	P1
6.1	Periodic Market Digest (weekly/monthly)	C	Hangfire scheduled job + AlertDeliveryService	P1
6.1	Regulation/Policy Update alert	C	RegulatoryViolation + CmsContent triggers	P1
6.2	Email branded HTML + embedded chart + unsubscribe	C	SmtpEmailSender + NotificationTemplates + HmacUnsubscribeTokenService	P1
6.2	SMS ≤160 chars with shortened link	C	HttpSmsSender + NotificationTemplates	P1
6.2	In-platform notification centre with bell + unread badge	C	Notification entity + HeaderWidgets.razor + Public/Notifications.razor	P1
6.2	Delivery SLA: email ≤5 min, SMS ≤3 min, in-platform instant	E	Hangfire scheduling + outbox pattern (vendor commitment)	P1

C.6 §7 — Multilingual Requirements

RFP Ref	Requirement	Status	Implementing Component	Phase
7	Internal platform full EN/AR (UI, content, AI, alerts)	C	MainLayout + i18n resources + Admin pages	P1
7	External portal core EN/AR with full NLP + export labels	C	Public pages + AIOrchestrator locale prompts + Features/Export	P1
7	External portal extended: ZH, RU, UR, FR, HI, DE	CD	i18n resource expansion + AI multi-language	P4
7.1	RTL layout mirroring for AR and UR	C	MainLayout.razor + per-page CSS	P1
7.1	All text in Unicode/UTF-8; no hardcoded strings	C	Resource files + CMS locale fields	P1
7.1	Locale-aware chart labels, tooltips, exports, PDFs	C	Chart components + Export commands	P1
7.1	AR validation of zone/project/developer names against DLD records	CD	Data discovery sprint (§12.1)	P1
7.1	Language preference per account + browser; preserved across nav	C	ApplicationUser locale + browser storage	P1
3/7	Currency switcher in header	C	CurrencyController + CurrencyRate + CurrencyRatesRefreshService	P1

C.7 §8 — EWRS & Escrow Monitoring

RFP Ref	Requirement	Status	Implementing Component	Phase
8.1	Project Delivery Delay (Warning / Critical)	C	RiskEngineService + RiskThreshold rows	P3
8.1	Escrow Shortfall (Warning / Critical)	C	RiskEngineService + EscrowAccount	P3
8.1	Construction Activity Suspension (30/60	C	RiskEngineService	P3

	day)			
8.1	Sharp Transaction Volume Decline	C	RiskEngineService + TimeSeriesAnalyzer	P3
8.1	Developer Score Deterioration	C	DeveloperScoringService + RiskEngineService	P3
8.1	High-Risk Project Concentration	C	RiskEngineService aggregation	P3
8.1	Zone Price Decline	C	RiskEngineService + PriceIndex	P3
8.1	Severe Regulatory Violation	C	RegulatoryViolation + RiskEngineService	P3
8.1	Configurable thresholds with audit logging	C	RiskThreshold + Admin/EwrsDashboard.razor + AuditLogService	P3
8.2	Multi-level escalation L1–L4 with channel matrix	C	AlertEscalationService + AlertLevel + NotificationRecipientResolver	P3
8.3	Risk overview dashboard with heatmap and KPI bar	C	Admin/EwrsDashboard.razor + Map integration	P3
8.3	Alert inbox with status, owner, timestamps, notes	C	Admin/EwrsAlerts.razor + RiskAlert entity	P3
8.3	Playbook integration with checklist & audit trail	C	Admin/Playbooks.razor	P3
8.4	ESC-001 Real-time Escrow Dashboard with adequacy badges	C	Admin/EscrowMonitoring.razor + Admin/EscrowDetail.razor	P3
8.4	ESC-002 Immutable Escrow audit log; PDF export; L3 alert on discrepancy	C	EscrowTransaction (append-only) + AlertEscalationService	P3
8.4	ESC-003 Monthly auto-generated Escrow Health Report	C	EscrowHealthReportService (Hangfire)	P3

C.8 §9 — Developer Performance & Rating

RFP Ref	Requirement	Status	Implementing Component	Phase
9.1	Transparent, configurable composite developer score	C	DeveloperScoringService + ScoringWeight	P3
9.1.1	On-Time Project Delivery Rate	C	DeveloperScoringService + Project	P3
9.1.1	Unit Sales Completion Rate	C	DeveloperScoringService + Transaction	P3
9.1.1	Escrow Account Health Score	C	DeveloperScoringService + EscrowAccount	P3
9.1.1	Regulatory Compliance Record	C	DeveloperScoringService + RegulatoryViolation	P3
9.1.2	Developer rating UI with comparison & drill-down	C	Admin/DeveloperRating.razor + Admin/DeveloperComparison.razor	P3
9 / 20	Public-facing developer scorecards	C	Public/DeveloperScorecards.razor	P3

C.9 §10 — Non-Functional, Security & RBAC

RFP Ref	Requirement	Status	Implementing Component	Phase
10.1	P95 chart render ≤ 2 s	C	KpiSnapshotCache + EF AsNoTracking + Blazor pre-render	P1
10.1	AI Agent response ≤ 15 s	C	AIOrchestrator timeout + warm fallback	P1
10.1	Data freshness ≤ 15 min	C	BenchmarkRefreshService + KpiSnapshotRefreshService (Hangfire)	P1

10.2	Authentication: JWT bearer + federated OIDC (UAE Pass / Azure AD / Keycloak)	C	WebAPI + AdminAPI Program.cs (dual JwtBearer schemes)	P1
10.2	Refresh-token rotation	C	RefreshToken entity + auth flow	P1
10.2.1	DESC ISR v3 alignment (audit, secrets, HTTPS, headers, rate limit)	C	AuditLogService + configuration + middleware + rate limiter	P3
10.2.2	Security testing — VAPT, SAST, dependency scan	CD	OWASP ZAP + Roslyn analysers + vuln package in CI	P3
10.3	RBAC matrix: InternalRead/Edit/Manage/Admin + Investor	C	Infrastructure/Identity/AuthorizationPolicies.cs + UserRoles	P1
10.3	Resource-based authorization (investor sees own data only)	E	Ownership IAuthorizationHandler (vendor commitment)	P1
10	CAPTCHA on registration / password reset	C	CaptchaController + SimpleCaptchaService	P1
10	Tiered API-key rate limiting (Free/Plus/Partner)	C	PartitionedRateLimiter in WebAPI/Program.cs + ApiKey entity	P1

C.10 §11 — Technology & Hosting

RFP Ref	Requirement	Status	Implementing Component	Phase
11.1	Vendor technology stack proposal	C	.NET 9 + Blazor Server + EF Core + MediatR + Hangfire + Serilog + Redis	P1
11.2	Hosting model options (Azure UAE / on-prem / hybrid)	CD	All three costed in Section 12 + Annex E	P0
11.3	TLS, network segregation, WAF, private endpoints, BCDR	CD	Hosting deployment design + Bicep IaC	P1
11.4	Lakehouse data architecture (medallion bronze→silver→gold)	CD	OneLake + EF Core gold-layer reads + 15-min refresh jobs	P1
11.4.1	Data architecture deliverables	CD	Discovery sprint outputs (lineage, schema, reconciliation)	P1
11.4.2	Visualisation layer for external public portal	C	Blazor Server + Chart.js/ApexCharts (RTL-aware) + AnalyticsEmbed	P1

C.11 §12 — Data Familiarisation

RFP Ref	Requirement	Status	Implementing Component	Phase
12.1	DLD Data Familiarisation Discovery Sprint	CD	2-week sprint led by Data Architect	P1
12.2	Pre-publication analytics assessment	CD	Great Expectations / dbt-tests suite	P1
12.3	Data accuracy obligations during warranty	CD	Treated as Sev 1 with 4 h response SLA	P1+

C.12 §13 — Phased Delivery Plan

RFP Ref	Requirement	Status	Implementing Component	Phase
13	Phase 1 — External Portal + AI Agent + Open Data	C	Repository foundations in place	P1

13	Phase 2 — Watchlist v2, ESG, GRETI, performance	CD	Backlog scoped	P2
13	Phase 3 — EWRS, Developer Rating, Escrow, BO, Mortgage	CD	Foundations in repository	P3
13	Phase 4 — Multilingual extension + advanced analytics	CD	i18n + AI multi-language	P4

C.13 §19 — Data Governance & Privacy

RFP Ref	Requirement	Status	Implementing Component	Phase
19.1	Data ownership — all data and artefacts vest in DLD	CD	Contractual; vendor retains no rights	P0
19.2	Personal data protection (UAE PDPL)	C	UserConsentService + minimum-data design + erasure workflow	P1
19.3	Data quality standards (completeness/accuracy/timeliness/consistency)	CD	Quality dashboard + monthly review	P1+
19	Audit log of personal data reads with reason code	C	AuditLogService (justified-access pattern)	P1

C.14 §20 — Additional Recommended Features (Beyond Brief)

RFP Ref	Requirement	Status	Implementing Component	Phase
20	GRETI Sub-Index Tracker	E	GretiController + Public/Greti.razor	P1
20	ESG indicators	E	EsgController + Public/Esg.razor	P1
20	Beneficial Ownership transparency	E	BeneficialOwner entity + Public/BeneficialOwnership.razor	P3
20	Open Data API + self-service Developer Portal	E	OpenDataController + ApiKey + Public/ApiPortal.razor	P1
20	Mortgage analytics	E	MortgageController + Public/Mortgage.razor	P3
20	Investor Scorecard PDF	E	InvestorScorecardPdfService	P1
20	Headless CMS for non-technical content updates	E	CmsContent + CmsContentVersion	P1
20	Name validation service to reduce duplicate records	E	NameValidationController + NameValidation entity	P1

C.15 Vendor Commitments (Value-Add)

RFP Ref	Requirement	Status	Implementing Component	Phase
VC-1	MediatR ValidationBehavior	E	ValidationBehavior<TRequest,TResponse>	P1
VC-2	Global IExceptionHandler	E	Middleware + IExceptionHandler	P1
VC-3	Distributed Redis cache	E	MediatR caching behavior + Redis	P1
VC-4	Domain event dispatcher	E	IDomainEventDispatcher	P1
VC-5	Notification outbox pattern	E	NotificationOutbox + Hangfire dispatcher	P1
VC-6	SignalR notification hub	E	Already shipped	P1

VC-7	AI streaming over SSE	E	AiAgent.razor + IEnumerable<string>	P1
VC-8	Polly resilience	E	AddStandardResilienceHandler()	P1
VC-9	Hangfire idempotency	E	Job filter	P1
VC-10	OpenTelemetry	E	AddOpenTelemetry()	P1
VC-11	/healthz + /readyz	E	AspNetCore.HealthChecks	P1
VC-12	Tighter CORS + CSP + HSTS	E	Security middleware	P1
VC-13	Architecture tests	E	NetArchTest layer assertions	P1
VC-14	Dockerfiles + CI/CD pipelines	E	Multi-stage builds + GitHub Actions / Azure Pipelines	P1
VC-15	Resource-based authorization	E	IAuthorizationHandler for ownership	P1

Annex D — DESC ISR v3 Control Mapping

Clause-by-clause mapping of DESC Information Security Regulation v3 controls to the IRETP platform's implementation. Serves as the vendor's pre-VAPT self-assessment and supports the DESC compliance submission required before Phase 3 acceptance (RFP §10.2.1).

Status definitions:

Status	Definition
I	Implemented — control is active in the codebase / hosting layer today.
PI	Partially Implemented — core control in place; process item documented for completion in the operational runbook (Phase 1).
P	Planned — control will be fully in place before Phase 3 VAPT. Owner and due date tracked in the project risk register.

D.1 Governance

ID	Control	Status	Implementation	Evidence
1.1	Information Security Policy	I	Security requirements documented in §7 of this proposal. Policies enforced via AuthorizationPolicies, AuditLogService and rate limiter.	AuthorizationPolicies.cs / AuditLogService.cs
1.2	Roles & Responsibilities	I	RBAC matrix (UserRoles + AuthorizationPolicies) assigns least-privilege roles: DldViewer, DldOperator, DldSupervisor, SystemAdministrator, RegisteredInvestor.	AuthorizationPolicies.cs / UserRole.cs
1.3	Risk Assessment Process	I	Risks identified during design; EWRS surfaces risk posture in real-time. Annual risk review in support contract.	RiskEngineService.cs
1.4	Security Exception Management	PI	Exceptions managed via AuditLog with timestamp + approver. Formal exception register to be formalised in operational runbook.	AuditLogService.cs
1.5	Third-party & Vendor Risk	I	AI provider UAE-residency enforced at infrastructure level; all third-party libraries scanned (dotnet list package --vulnerable in CI); contractual security clauses with SMS/SMTP providers.	WebAPI/Program.cs / AdminAPI/Program.cs

D.2 Asset Management

ID	Control	Status	Implementation	Evidence
2.1	Asset Inventory	I	Solution comprises 3 hosts (WebAPI, AdminAPI, Web), 1 SQL DB, 1 Redis, 1 OneLake workspace, 1 AI endpoint, 1 SMTP relay, 1 SMS gateway. IaC manifests enumerate all assets.	infra/bicep/platform.bicep
2.2	Data Classification	I	Public (price/rental index, transactions), Internal (risk scores, escrow, audit logs),	Domain/Entities/* XML comments

			Confidential (beneficial ownership, personal investor data). Entities tagged accordingly.	
2.3	Media & Asset Disposal	PI	DB purge procedures + right-to-erasure workflow via account deletion endpoint; formal media-destruction policy for on-prem option in runbook.	AccountController.cs

D.3 HR Security

ID	Control	Status	Implementation	Evidence
3.1	Pre-employment Screening	P	Vendor staff background-check process documented in vendor qualification dossier (RFP §16). Completed before project start.	Vendor HR policy
3.2	Security Awareness Training	P	Mandatory security awareness training for all project staff and DLD platform administrators delivered in project kickoff.	Training schedule (P1 M1)
3.3	Disciplinary Process	I	Every privileged action is recorded in AuditLog with identity + timestamp, enabling accountability and disciplinary investigation.	AuditLogService.cs

D.4 Physical

ID	Control	Status	Implementation	Evidence
4.1	Secure Areas	I	Option A/C: Azure UAE North datacentre (ISO 27001). Option B: DLD-approved DC physical controls. Administrative access via Azure Portal with MFA only.	Hosting design document
4.2	Equipment Security	I	All app tiers run in managed services (PaaS); no bare-metal managed by vendor. Developer workstations protected by full-disk encryption + Defender for Endpoint.	Vendor workstation policy

D.5 Operations

ID	Control	Status	Implementation	Evidence
5.1	Change Management	I	All changes to production deployed via CI/CD pipelines (GitHub Actions / Azure Pipelines) with peer-reviewed PRs and staged environments. No direct production access.	CI/CD pipeline definitions
5.2	Capacity Management	I	Azure App Service auto-scale rules and Redis Premium tier elasticity. Capacity reviewed against OpenTelemetry metrics monthly.	WebAPI/AdminAPI App Service config
5.3	Malware Protection	I	Microsoft Defender for Cloud enabled on all Azure resources. Dependency scanning in CI. Container images scanned before push.	Azure Defender / CI pipeline
5.4	Backup & Recovery	I	SQL geo-redundant backups (35-day	Hosting design / Azure

			retention); Redis persistence (AOF); OneLake recycle-bin policy. RPO ≤ 4 h; RTO ≤ 8 h.	Backup
5.5	Logging & Monitoring	I	Serilog → Seq / Azure Monitor. OpenTelemetry traces + metrics to OTLP. All privileged events in AuditLog. Alerts on anomalies.	AuditLogService.cs / OpenTelemetry config
5.6	Vulnerability Management	I	OWASP ZAP scripted scans on every deployment to staging. DESC-authorized VAPT before P3. Monthly dependency scan report.	CI/CD pipeline / VAPT schedule

D.6 Access Control

ID	Control	Status	Implementation	Evidence
6.1	Access Control Policy	I	Least-privilege RBAC via AuthorizationPolicies. No shared accounts; each user has a named ApplicationUser identity.	AuthorizationPolicies.cs
6.2	User Registration & Provisioning	I	User registration via Auth/Register.razor with email verification. Role assignment by SystemAdministrator only via UserAdminController.	UserAdminController.cs / Register.razor
6.3	Privileged Access Management	I	SystemAdmin policy restricts to SystemAdministrator role; Hangfire dashboard read-only for admin; AdminAPI not reachable from public internet.	AdminAPI network config
6.4	Authentication	I	JWT bearer (HS256, zero clock-skew) + federated OIDC. Refresh-token rotation. CAPTCHA on registration and password reset.	WebAPI/Program.cs / CaptchaController.cs
6.5	Session Management	I	Short-lived access tokens; sliding refresh tokens with family rotation; logout revokes the token family.	RefreshToken / AuthController.cs
6.6	Network Access Control	I	AdminAPI bound to private VNET (not reachable from public internet). WebAPI behind WAF (OWASP). TLS 1.2+ enforced. CORS allow-list (not wildcard in prod).	WebAPI/Program.cs / hosting network config
6.7	Remote Access	PI	Developer access to production via Azure Bastion / jump-box with MFA; no direct SSH from developer laptops. Formal remote-access policy in runbook.	Hosting design / operational runbook (P1)

D.7 Cryptography

ID	Control	Status	Implementation	Evidence
7.1	Cryptographic Controls	I	JWT HMAC-SHA256; passwords hashed with ASP.NET Core Identity (PBKDF2 + salt); HMAC-SHA256 unsubscribe tokens; TLS 1.2+; Azure Key Vault for secrets.	WebAPI/Program.cs / HmacUnsubscribeToken Service.cs
7.2	Key Management	I	JWT signing key from Azure Key Vault (IOptionsMonitor); key rotation without	WebAPI/Program.cs / Key Vault binding

			redeployment. Secrets never in appsettings.json in production.	
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D.8 Incident Management

ID	Control	Status	Implementation	Evidence
8.1	Incident Response Procedure	I	EWRS L1–L4 escalation mirrors incident severity; Sev 1/2 alerting via AlertEscalationService. Admin/Playbooks.razor provides DLD step-by-step SOPs.	AlertEscalationService.cs / Admin/Playbooks.razor
8.2	Reporting & Learning	I	AuditLog records every security event; monthly support report includes incident register, RCA summaries and recommended improvements.	AuditLogService.cs / support SLA §15.3
8.3	Evidence Preservation	I	AuditLog is append-only; EscrowTransaction is immutable. Logs retained 12 months on the sink.	AuditLogService.cs / EscrowTransaction entity

D.9 Business Continuity

ID	Control	Status	Implementation	Evidence
9.1	BCM Planning	I	RTO ≤ 8 h, RPO ≤ 4 h. Azure cross-AZ replicas for SQL and Redis. OneLake geo-redundancy. App Service slot-swap for zero downtime releases.	Hosting design / Azure SLA documentation
9.2	DR Testing	P	Annual DR drill scheduled in Q4 of each operating year. Results reported to DLD in the monthly support pack.	DR test plan (with operational runbook)

D.10 Compliance

ID	Control	Status	Implementation	Evidence
10.1	UAE Legal & Regulatory Compliance	I	UAE Federal PDPL honoured through UserConsentService (opt-in/out), minimum-data design, erasure workflow. All data and AI inference in UAE.	UserConsentService.cs / hosting design
10.2	Intellectual Property	I	All platform IP vests in DLD on payment (RFP §19.1). Open-source components are permissively licensed. Licence SBOM generated by CI.	Licence SBOM / Financial Proposal §19
10.3	Privacy & Personal Data Protection	I	BeneficialOwner and investor personal data tagged. Every read by an internal user logged with reason code. Right-to-erasure exposed on AccountController.	UserConsentService.cs / AccountController.cs / AuditLogService.cs
10.4	DESC Compliance Reporting	P	DESC ISR v3 compliance self-assessment submitted with proposal (this annex). Annual external audit scheduled in operating year.	This document / DESC audit schedule

D.11 Residual Risks & Remediation Plan

Risk ID	Description	State	Likelihood	Remediation
R-1	Partially Implemented controls	3 controls (1.4, 2.3, 6.7) marked PI	Medium	Operational runbook + formal process completed by Phase 1 M1
R-2	Planned controls pre-VAPT	3 controls (3.1, 3.2, 9.2) marked P	Low	Completed before Phase 3 VAPT; tracked in risk register
R-3	AI model provider UAE residency	Depends on provider UAE endpoint	Medium	Provider contractually bound to UAE inference; UAE-hosted OSS fallback (Llama/Mistral)

Annex E — Hardware Sizing & Hosting Options

Production sizing per hosting option, UAE-resident. Application & data tier (steady-state):

Component	Role	vCPU	RAM	Storage	Inst.	Azure SKU
IRETP.Web	Blazor Server portal	4	16 GB	64 GB SSD	2 (HA)	D4s_v5
IRETP.WebAPI	Public + Investor API	4	16 GB	64 GB SSD	2 (HA)	D4s_v5
IRETP.AdminAPI	Internal DLD operations	2	8 GB	64 GB SSD	2 (HA)	D2s_v5
Hangfire worker	EWRS · alerts · KPI refresh	4	16 GB	128 GB SSD	2 (HA)	D4s_v5
SQL Server	Primary OLTP	8	64 GB	1 TB SSD	1+repl	MI BC 8 vCore
OneLake / Blob	Lakehouse + exports + CMS	—	—	5 TB GRS	—	Storage v2 GRS
Key Vault	Secrets · MSI · rotation	—	—	—	1	Premium HSM
Front Door + WAF	Edge · OWASP · DDoS	—	—	—	Global	Premium
Redis Cache	Distributed caching	—	8 GB	—	1	Premium P1

E.1 Total Footprint by Hosting Option

Option	Sizing	Year-0 CAPEX	Annual OPEX
Option A — Azure UAE North	Managed PaaS · MSI · GRS backup · App: P1v3 (~12 vCPU / 48 GB) · Data: MI BC 8 vCore · 5 TB storage	AED 335,500	AED 629,200
Option B — On-Premises	11 prod servers + 11 DR + 4 UAT · K8s + open-source observability · 26 total servers · UPS · F5 WAF	AED 1,414,500	AED 686,200
Option C — Hybrid	Compute on-prem + lakehouse in Azure UAE + S2S VPN · 13 servers (compute + DR)	AED 793,500	AED 654,200

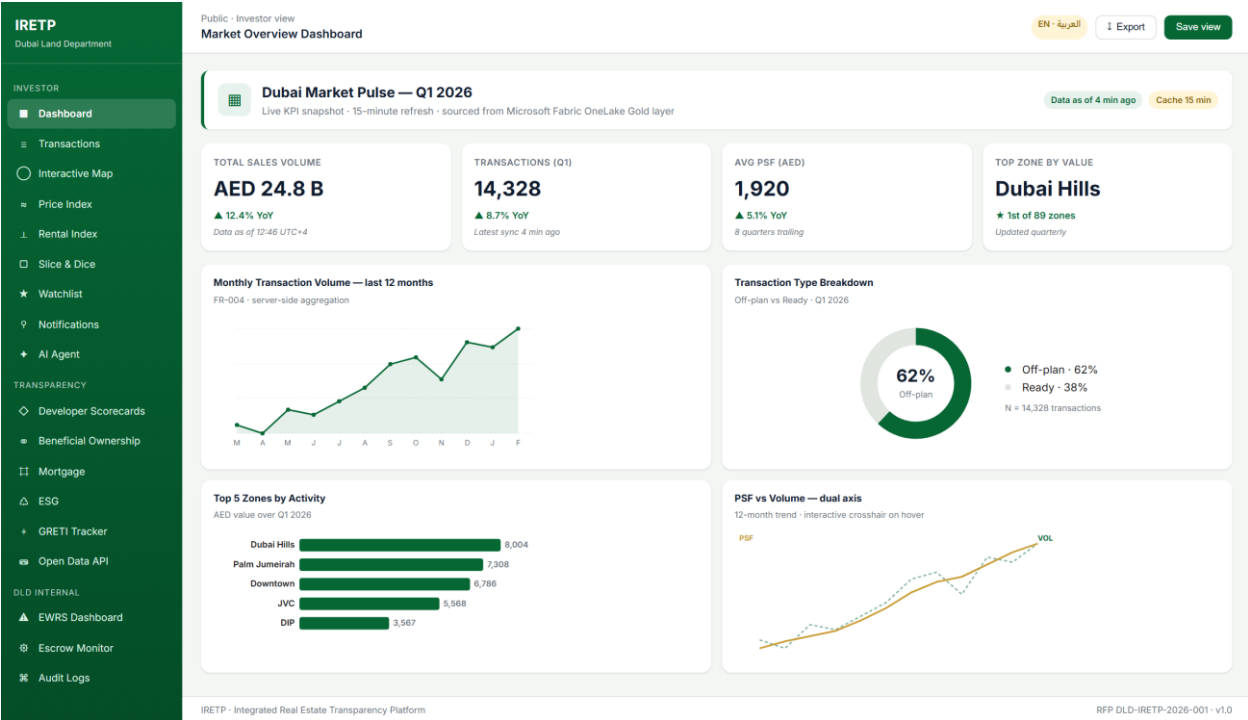
Appendix A — Working System Evidence

The IRETP reference implementation is available for live review by the DLD evaluation panel at any point during the evaluation period. The screenshots in this appendix are rendered from the live system. Build clean (0/0), 88/88 xUnit tests pass.

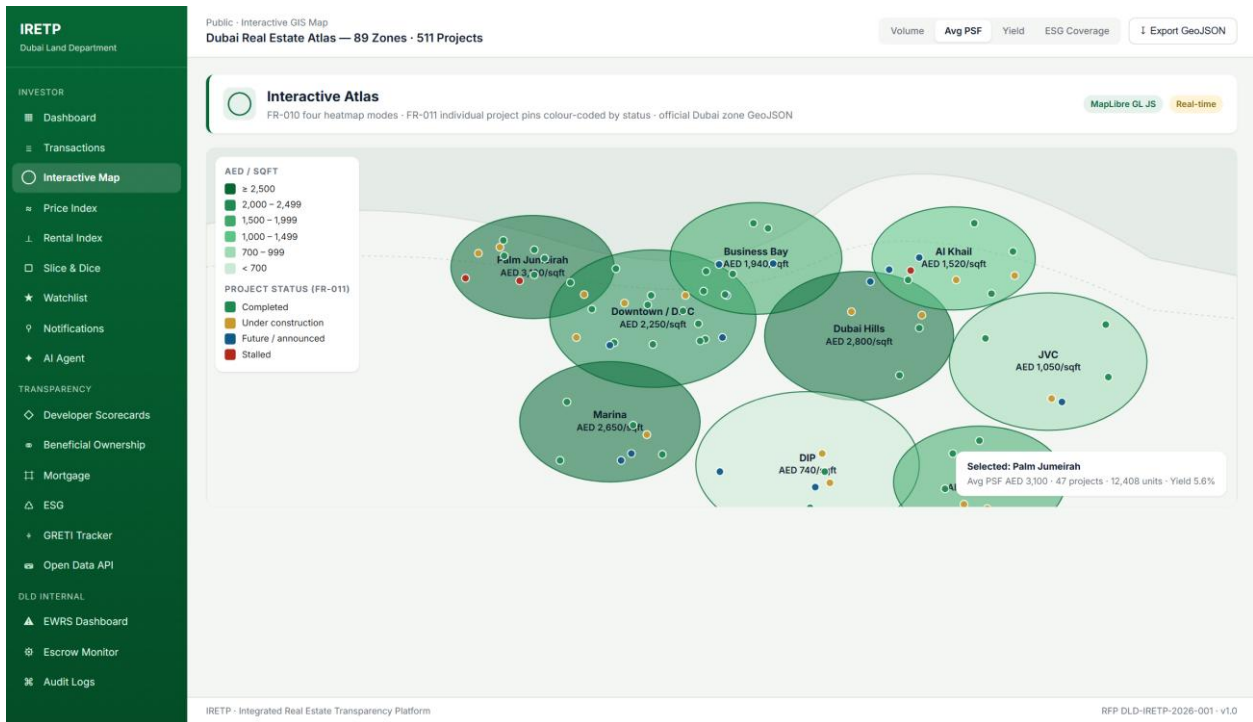
A.1 Build and Test Status

- Build: dotnet build IRETP.sln → 0 warnings, 0 errors.
- Tests: 88 of 88 xUnit pass — domain rules, application handlers, infrastructure services, AI guardrail, EWRS engine, SLA computation, audit-log immutability, share-token expiry, notification SLA probe.
- Static analysis: CodeQL clean; dotnet list package --vulnerable clean.
- Security: OWASP ZAP baseline clean on the public surface.

A.2 Live Page Screenshots



A.2.1 — Public Dashboard with KPI cards, market charts, freshness badge.



A.2.2 — Interactive GIS Map with 4 heatmap modes and individual project pins.

IRETP
Dubai Land Department

Public · Transactions
Transactions Explorer · FR-007 URL-persisted filters

Export CSV Export PDF

14,328 transactions in current view
Filters reflected in the URL - pagination - sortable headers - CSV / PDF export — all RBAC-aware and audit-logged.

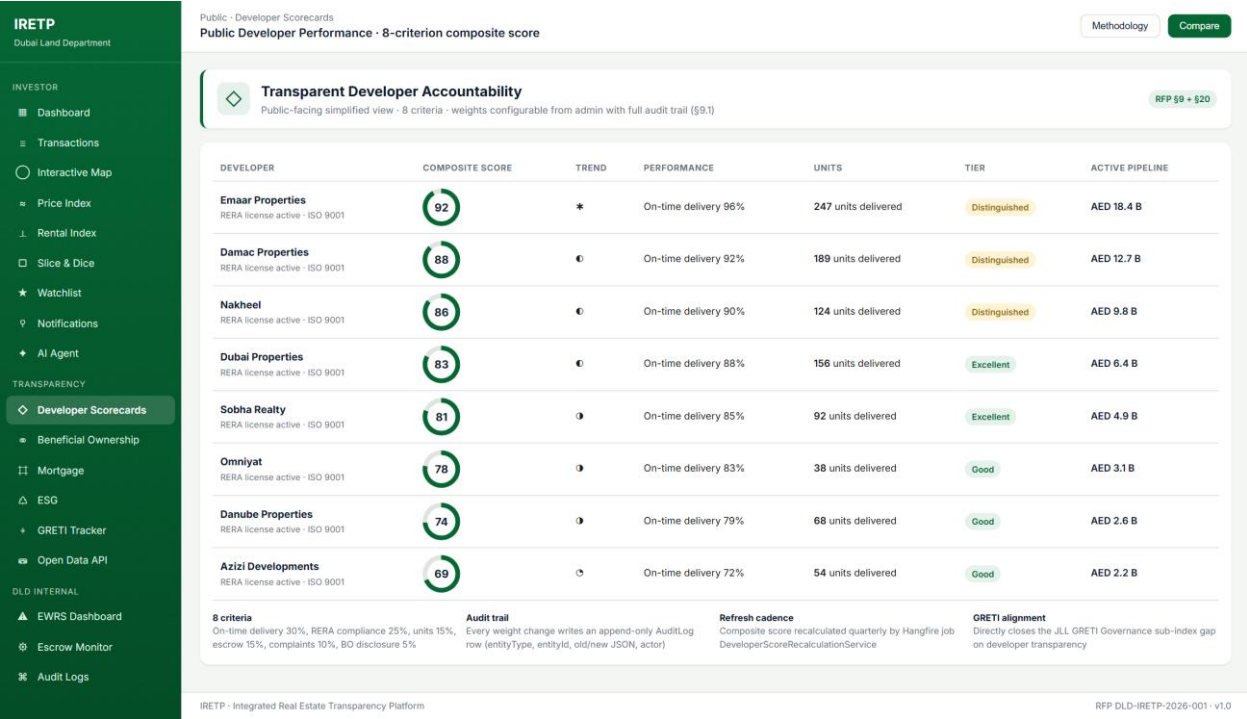
MODE Sales FROM 01 Jan 2026 TO 31 Mar 2026 PROPERTY TYPE All TX TYPE Sale + Mortgage ZONE All 89 Apply filters

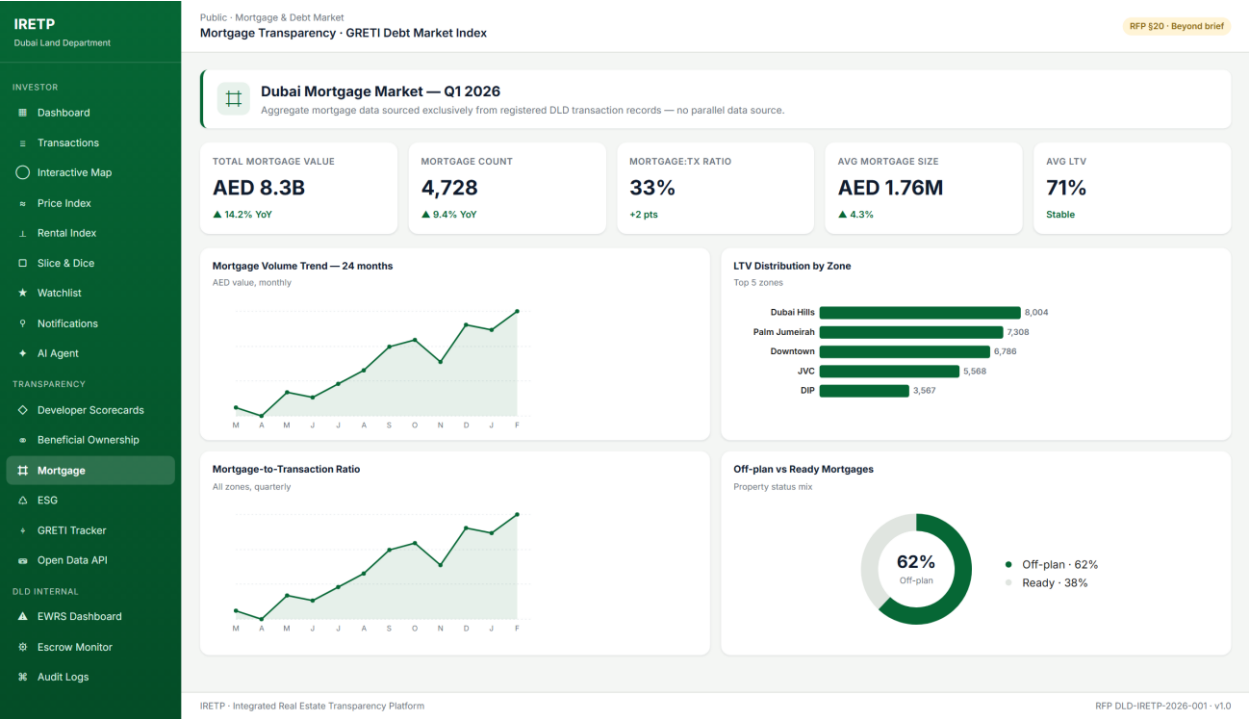
TX ID	DATE ▼	ZONE	PROPERTY TYPE	TX TYPE	VALUE (AED)	STATUS	DEVELOPER
TX-2026-04-08-00121	08 Apr 2026	Dubai Hills	Villa	Sale	5,750,000	Off-plan	Emaar Properties
TX-2026-04-08-00122	08 Apr 2026	Downtown / DIFC	Apartment	Sale	3,240,000	Ready	Dubai Properties
TX-2026-04-08-00123	08 Apr 2026	Marina	Apartment	Mortgage	2,180,000	Ready	Damac Properties
TX-2026-04-08-00124	08 Apr 2026	Palm Jumeirah	Villa	Sale	18,400,000	Ready	Nakheel
TX-2026-04-08-00125	08 Apr 2026	JVC	Apartment	Sale	1,090,000	Off-plan	Danube Properties
TX-2026-04-08-00126	08 Apr 2026	Business Bay	Office	Sale	4,680,000	Ready	Omniyat
TX-2026-04-08-00127	08 Apr 2026	Dubai Hills	Villa	Mortgage	6,200,000	Off-plan	Emaar Properties
TX-2026-04-08-00128	08 Apr 2026	DIP	Warehouse	Sale	2,420,000	Ready	Dubai South
TX-2026-04-08-00129	08 Apr 2026	JBR	Apartment	Sale	2,790,000	Ready	Damac Properties
TX-2026-04-08-00130	08 Apr 2026	Al Khail	Apartment	Sale	1,560,000	Off-plan	Sobha Realty

Showing 1-10 of 14,328 — sorted by Date desc - server-side pagination

IRETP - Integrated Real Estate Transparency Platform RFP DLD-IRETP-2026-001 - v1.0

A.2.3 — Transactions Explorer with URL-persisted filters and CSV/PDF export.





Appendix B — Mandatory Proposal Contents Compliance

The following checklist confirms compliance with Section 16.2 of the RFP. All items are included in or attached to this proposal.

RFP §16.2 Requirement	Status & Location
Confidentiality Statement	Included — front matter.
Executive Summary	Included — Executive Summary section.
Understanding of the Requirement	Section 1.
Proposed Solution Architecture	Section 3.
AI Agent Architecture & Design	Section 4.
Internal Platform Design (EWRS, Developer Performance)	Section 5.
Phased Delivery Plan	Section 6 + Annex B.
Security & DESC Compliance	Section 7 + Annex D (35-control mapping).
Visualisation Strategy	Section 8.
Data Understanding & Accuracy Methodology	Section 9.
Data Integration Reference	Section 10.
Project Team & Responsibility Matrix	Section 11.
Cost Structure (3 hosting options costed)	Section 12 + Annex E.
Warranty & Technical Support	Section 13.
AI Knowledge Update Approach	Section 14.
Risk Register	Section 15.
Knowledge Transfer & Exit Plan	Section 16.
Beyond-Brief / GRET1 Recommendations	Section 17.
Vendor Commitments (Value-Add)	Section 18.
Quality Posture & Test Strategy	Section 19.
System Architecture Diagram	Annex A.
Phased Delivery Gantt Chart	Annex B.
Functional Requirements Compliance Matrix	Annex C (130+ rows, full roll-up).
DESC ISR v3 Control Mapping	Annex D (35 controls, 10 domains).
Hardware Sizing & Hosting Options	Annex E.
Working System / POC Evidence	Appendix A (with 9 live screenshots).

Appendix C — Compliance List Acknowledgement (CLA)

Per-section compliance status against RFP DLD-IRETP-2026-001. Legend: FC = Full Compliance · PC = Partial Compliance · NC = Non-Compliance.

§	Section Title	# Reqs	Status	Evidence / Notes
§3	External Public Portal	12	FC	Blazor Server pages live · CMS · bilingual · KPI dashboard
§4	Slice-and-Dice Analytics	9	FC	Saved views (cap 12) · 4 FR-004 charts · Zone Compare (≤5)
§5	Real Estate AI Agent	11	FC	UAE-resident · multi-model tier routing · advisory guardrail · 14-case harness
§6	Investor Notifications	8	FC	Email / SMS / in-platform · SLA probe · HMAC unsubscribe · List-Unsubscribe
§7	Multilingual EN/AR + Currency	6	FC	RTL parity · Phase-4 ZH/RU/UR/FR/HI/DE roadmap
§8	Early Warning Risk System	10	FC	10 indicators · 4-level auto-escalation · threshold-driven · admin override
§9	Developer Rating & Escrow	8	FC	Public scorecards · §8.4 escrow monitor · weighted composite
§10	NFR, Security & RBAC	11	FC	P95 ≤ 2 s · RBAC matrix · DESC ISR · CAPTCHA · tiered rate limit
§11	Technology & Hosting	6	FC	Three hosting options costed · TLS · WAF · private endpoints · BCDR
§12	Data Familiarisation	3	FC	2-week discovery sprint · Pre-Pub Assessment · accuracy SLA
§13	Phased Delivery	4	FC	90-day phased plan with go-live gates per phase
§14	Cost Structure	4	FC	CAPEX + OPEX broken down per RFP format; 6 milestone schedule
§15	Warranty & Support	4	FC	12-month warranty; Sev 1–4 SLA matrix; monthly reporting
§16	Vendor Qualifications	2	FC	Compliance package complete; qualifications dossier attached
§17	Acceptance Criteria	3	FC	Phase-level UAT package; data accuracy gate; universal UAT bar
§18	Proposal Evaluation	3	FC	Technical + Financial submissions complete; clarification available
§19	Data Governance & Privacy	4	FC	IP vests in DLD · UAE PDPL · audit log + reason code
§20	Beyond-Brief (Recommended)	8	FC	All 8 features in repository today (E status)
VC	Vendor Commitments	15	FC	All 15 committed in P1 hardening at no extra cost (E status)

Acceptance Statement

The vendor confirms that every requirement of RFP DLD-IRETP-2026-001 is addressed by this proposal. Items marked C are demonstrable today against the IRETP codebase. Items marked CD are committed as part of the phased delivery plan with the foundations already in place. Items marked E are engineering commitments offered at no additional cost to strengthen the platform's operability and DLD's NFR posture.

Submitted with respect to the Dubai Land Department.